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ELECTROMAGNETIC ENVIRONMENTAL EFFECTS TO ORDNANCE GUIDE

PART 2

HAZARDS OF ELECTROMAGNETIC RADIATION TO ORDNANCE TESTING

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FOREWORD

1. This handbook is approved for use by all Departments or Agencies of the Department of Defense (DoD).

2. This part supplements MIL-HDBK-240-1 and was prepared for contractors, DoD test agencies, activities, and associated stakeholders. This handbook is for guidance only and cannot be cited as a requirement.

3. This part of the handbook provides recommended practices for conducting Hazards of Electromagnetic Radiation to Ordnance (HERO) evaluations. Developed over many years by DoD test activities, the methods and procedures described herein are the product of the collective DoD HERO test experience. This handbook consolidates these practices into a single document vice numerous individual Service standards, instructions, and so forth, thus promoting a standardized approach.

4. This handbook was prepared by the Services under the sponsorship of the Joint Ordnance Commander's Group (JOCG) Electromagnetic Environmental Effects (E3) Ordnance Safety Subgroup in accordance with the guidelines of the Standardization Reform Policy established by the Secretary of Defense.

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CONTENTS

PARAGRAPH		PAGE
FOREWORD		ii
1.	SCOPE	1
1.1	Purpose.	1
1.2	Background.	1
2.	APPLICABLE DOCUMENTS	1
2.1	General.	1
3.	DEFINITIONS	1
3.1	Definitions.	1
4.	HERO TEST PROCESS	2
4.1	Overview.	2
4.2	Ordnance Documentation.	2
4.2.1	Mission.	2
4.2.2	Ordnance Description.	2
4.2.3	Associated Equipment.	3
4.2.4	S4 Description.	3
4.3	Pre-test Assessment.	4
4.3.1	EID Description and Characteristics.	5
4.3.1.1	Types of EIDs.	5
4.3.1.2	Firing Sensitivity.	7
4.3.1.3	Bridgewire Resistance.	7
4.3.1.4	Response Time, τ.	7
4.3.1.5	Thermal Stacking.	8
4.3.1.6	EID Function.	9
4.3.1.7	Firing Consequence.	9
4.3.2	Evaluation of RF-Protective Features and Weaknesses.	10
4.3.3	Enclosures.	10
4.3.3.1	Cables and Connectors.	10
4.3.3.2	Electromagnetic Interference (EMI) Gaskets.	11
4.3.3.3	EMI Filters.	11
4.3.4	Firing Systems.	11
4.3.4.1	Typical EID Circuit Design.	12
4.3.4.2	Energized vs. Non-Energized Firing Circuits.	14
4.3.4.3	In-Line vs. Out-of-Line Circuits/Explosive Trains.	14
4.3.5	Multiple Platform/System Analysis.	15
4.3.6	Response Prediction.	15
4.3.6.1	Results of Previous Tests.	15
4.3.6.2	Analytical Prediction of Resonances.	16
4.3.6.3	Analytical Tools.	16
4.3.6.4	Reverberation Chamber.	16
4.3.6.5	Ordnance Operating Frequencies.	17

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:39Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
4.3.6.6	Supplemental Tests.	17
4.4	Test Configuration.	18
4.4.1	Physical Configurations.	18
4.4.1.1	Ordnance SUT.	18
4.4.1.2	Ordnance Configurations.	19
4.4.1.3	Ancillary Equipment.	21
4.4.2	Operating Procedures.	22
4.4.2.1	Standard Operating Procedures (SOPs).	22
4.4.2.2	Non-Standard (Not Specifically Authorized) Operating Procedures.	22
4.4.3	Expected EME.	22
4.5	Test EME Levels.	22
4.5.1	EME Parameters.	23
4.5.1.1	Test Frequencies.	23
4.5.1.2	Field Strength Levels.	24
4.5.1.3	Illumination Angles.	25
4.5.1.4	Polarization.	26
4.5.1.5	Modulation.	26
4.6	Selection of a Test Facility.	27
4.6.1	OATS Facility.	28
4.6.2	Anechoic Chamber.	29
4.6.3	Reverberation Chamber	31
4.7	EID Instrumentation.	33
4.7.1	Instrumentation Characteristics.	33
4.7.1.1	Sensitivity.	34
4.7.1.2	Instrumentation System Response Time.	34
4.7.1.3	Effects of Instrumentation on SUT.	34
4.7.2	Types of Instrumentation.	35
4.7.2.1	Sensors and Transducers.	36
4.7.2.2	Actual EID vs. Substituted Devices.	37
4.7.2.3	Optical vs. Conductive-Type Sensors.	37
4.7.2.4	Sensor-to-Bridgewire Proximity.	37
4.7.2.5	Computers and Recording Devices.	37
4.7.3	Bridgewire EIDs and Ordnance Instrumentation.	38
4.7.4	Instrumentation/Monitoring Firing Circuits and Electronic Devices.	39
4.7.5	Non-Instrumented Testing (Go/No-Go Type).	39
4.7.5.1	Safety/Reliability Margin Verification Test.	39
4.7.5.2	Statistical Sample Size Verification Test.	40
5.	TEST PLAN	40
5.1	Introductory Information.	40
5.2	System Background.	40
5.3	Purpose.	41
5.4	Scope.	41

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:39Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
5.5	System Test Configuration.	41
5.5.1	Ordnance Item Description.	41
5.5.2	Ordnance Item Mission.	41
5.5.3	Description of Test Item's S4 Phases.	41
5.5.4	Ordnance Item Test Configuration.	42
5.6	Description of the HERO Test Facility.	42
5.7	Description of the Test Environments.	42
5.8	Test Instrumentation.	42
5.9	Compliance Criteria.	42
5.10	Test Schedule.	43
5.11	Safety Precautions.	43
6.	CONDUCTING THE TEST	43
6.1	Introduction.	43
6.2	Safety.	43
6.2.1	SOP.	43
6.2.2	Safety Briefings.	43
6.2.3	Ordnance Handling.	44
6.2.4	MPE.	44
6.2.5	Compliance with MPEs.	44
6.2.6	Operations Involving Host Platforms and Ancillary Equipment.	44
6.2.7	Personal Protective Equipment (PPE).	44
6.3	HERO Test EME.	45
6.4	General EME Calibration Procedures.	48
6.4.1	Free-field Calibration.	48
6.4.2	SUT Positioning.	49
6.4.3	Pre-test Platform Characterization Measurements.	50
6.5	Generating Test EMEs.	50
6.5.1	HF Environment (2-30 MHz).	50
6.5.1.1	Field Strength Levels.	50
6.5.1.2	Transmitting Antennas.	51
6.5.1.3	Field Calibration.	52
6.5.1.4	Illumination Angles/Illumination Area.	52
6.5.1.5	Polarization.	53
6.5.1.6	Modulations.	53
6.5.2	VHF Environment (30-225 MHz).	53
6.5.2.1	Field Strength Levels.	53
6.5.2.2	Transmitting Antennas.	53
6.5.2.3	Field Calibration.	54
6.5.2.4	Illumination Angles/Illumination Area.	54
6.5.2.5	Polarization.	55
6.5.2.6	Modulations.	55
6.5.3	Frequencies Above 225 MHz.	55

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:39Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
6.5.3.1	Field Strength Levels.	55
6.5.3.2	Transmitting Antennas.	56
6.5.3.3	Field Calibration.	56
6.5.3.4	Illumination Angles/Illumination Area.	57
6.5.3.5	Polarization.	57
6.5.3.6	Modulations.	57
6.6	Maximizing Responses.	57
6.6.1	Illumination Angles.	57
6.6.2	Frequency Increments.	57
6.6.3	Stimulus Dwell Time.	57
6.6.4	Communications.	58
6.7	Non-Linear Effects.	58
6.7.1	RF Arcing.	58
6.7.2	Energized Circuit Susceptibility.	58
6.8	Susceptibility Investigation.	59
6.8.1	Identifying POE.	59
6.8.2	HERO Fixes and Verification.	59
7.	DATA ANALYSIS	59
7.1	Introduction.	59
7.2	Detected Responses.	60
7.3	Undetected Responses.	60
7.4	Compliance Criteria (Required Margins).	60
7.4.1	Safety Firing Consequence Factors (FCFs).	60
7.4.2	Reliability FCFs.	60
7.5	Conversion to EID Response at Criteria EME Level.	61
7.5.1	Determination of Compliance.	61
7.5.2	Determination of Compliance Margin.	61
7.6	Conversion to MAE.	62
7.7	Extrapolation Limits.	62
7.8	Determination of MAE per Frequency Band.	62
8.	TEST REPORT	63
8.1	Introductory Information.	63
8.2	Main Body Sections.	63
8.2.1	Introduction.	63
8.2.2	Background.	63
8.2.3	Purpose.	63
8.2.4	Scope.	63
8.2.5	Test Results Discussion.	63
8.2.6	Conclusions.	63
8.2.7	Recommendations.	64
8.2.8	Appendices.	64

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:39Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
9.	ORDNANCE CHANGES THAT WOULD SIGNAL A RETEST	67
9.1	EID Changes That Require a HERO Test.	67
9.2	Firing Circuit Changes That Require a HERO Test.	68
9.3	Ordnance Envelope Changes That Require a HERO Test.	68
9.4	S4 Changes That Require a HERO Test.	68
10.	DOCUMENTATION	68
10.1	Introduction.	68
10.2	Purpose of Test Documentation.	68
11.	NOTES	69
11.1	Intended Use.	69
11.2	Supersession Data.	69
11.3	Cross-Reference of Classification and Substitutability Data.	69
11.4	Subject Term (Keyword) Listing.	69
11.5	Identification of Changes.	69
11.6	Technical Points of Contact.	69

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Check the source to verify that this is the current version before use.

CONTENTS

FIGURE		PAGE
1.	Stockpile-to-safe-separation sequence (S4)	4
2.	Illustration of thermal stacking of MK 1 squib during 9 GHz radar test	8
3.	Typical HBW firing circuit with an S&A device	12
4.	Differential mode of RF excitation in a two-wire firing system	12
5.	Coaxial mode RF excitation in a coaxial firing system	13
6.	Coaxial mode RF excitation in a two-wire firing system	14
7.	Example of Illumination Angles	26
8.	Typical OATS test facility with ground plane	28
9.	Typical anechoic chamber	30
10.	Typical reverberation chamber	31
11.	Metal-junction thermocouple instrumentation for bridgewire-type EIDs	36
12.	Typical Optical sensor instrumentation for bridgewire-type EIDs	36
13.	Bridgewire-type EID Instrumentation	37
14.	OATS	45
15.	EME transmission equipment and antennas block diagram	46
16.	HF field calibration technique	46
17.	VHF/UHF range calibration technique	47
18.	Radar range calibration technique	47
19.	SUT in front orientation with turret at 90° from an HF antenna	53
20.	Illumination of test item at VHF frequencies	55
21.	RF arc generated during HERO testing	58

TABLE		PAGE
I.	EID Identification/Characteristics Example	5
II.	Typical EID Characteristics	6
III.	Representative HERO Test Frequencies in MHz	24
IV.	HERO EME Test Levels	25
VI.	Sample Test Results Sheet Format	65
VII.	Sample HERO Summary Datasheet	66

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1. SCOPE

1.1 Purpose. This part of MIL-HDBK-240 provides the procedures by which the hazards of electromagnetic radiation to ordnance (HERO) margins defined in MIL-STD-464 are verified. It covers HERO testing for all Air Force, Army, Navy, Marine Corps, and Coast Guard ordnance and support equipment for all mission areas. Although intended primarily for use by Department of Defense (DoD) HERO test activities, this part of the handbook provides a consolidation of "corporate knowledge" about the subject that should be of interest to procurement authorities and system developers.

1.2 Background. There are four specific objectives of this handbook:

a. Document HERO Joint-Service test methodology.

b. Promote test standardization.

c. Identify alternative techniques and instrumentation.

d. Facilitate the exchange of HERO test data.

Ultimately, HERO test data is used to determine the maximum allowable environment (MAE) for ordnance and weapon systems containing electrically initiated devices (EIDs). MAE information is used to assess HERO risks and develop effective control measures to minimize those risks.

HERO assessments are essentially a determination of how EIDs respond to the electromagnetic environment (EME) that could be encountered by the ordnance throughout its service life. The response can be influenced by three principal factors: (1) EME characteristics, (2) physical configuration(s) of the ordnance, and (3) handling procedures. Each factor should be considered in the context of the stockpile-to-safe separation sequence (S4) (4.2.4).

When defining the operational EME, one should anticipate not only the individual Service EME, but also the EME encountered during Joint operations. The JCIDS Manual states that, "Because all weapons/weapon systems have the potential of being deployed together or employed in Joint environments, weapons and weapon systems will be considered Joint systems within the JCIDS process and may be designated as Joint Requirements Oversight Council (JROC) or Joint Capabilities Board (JCB) Interest." As Joint operations become more commonplace, there is an increased likelihood that weapons from one Service will be exposed to other Services' EME levels. MIL-STD-464 specifies the EME levels for HERO certification.

2. APPLICABLE DOCUMENTS

2.1 General. See MIL-HDBK-240-1. This document can be found on ASSIST Online at https://assist.dla.mil.

3. DEFINITIONS

3.1 Definitions. See MIL-HDBK-240-1.

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4. HERO TEST PROCESS

4.1 Overview. MIL-STD-464 establishes electromagnetic environmental effects (E3) interface requirements and verification criteria for systems, including associated ordnance. Both the efficiency with which a HERO verification test is conducted and the quality of the data produced depends largely on the effort that is put into the planning process. Proper test planning requires that certain key information be obtained about the test item. The process begins with a complete identification of the ordnance item, including all components/sub-components, host platform, and all EIDs contained within the ordnance. Additional crucial information includes mission profile, description of the ordnance S4 phases, and the system's physical characteristics. This information then is analyzed to determine how the test should be conducted; for example, which S4 configurations should be tested, what specific procedures are necessary to exercise the ordnance, and what test EME levels should be generated. In essence, the planning phase consists of gathering and evaluating information. An engineering evaluation of the information, with due consideration to test resource constraints, then should be made to develop a plan for accomplishing the test objectives in the most efficient manner. The following paragraphs provide guidance for developing such a test plan (see paragraph 5).

4.2 Ordnance Documentation. An important step in preparing for a HERO test is to gather certain key information; this is a prerequisite to all other steps in the preparation process. The minimum information needed is detailed below and should be included in the documentation package.

4.2.1 Mission. A description of the mission of the ordnance item is needed to understand how the item will be used throughout its service life and becomes the basis for determining appropriate HERO test procedures. This description is the basis for selecting the S4 configurations to be tested. Furthermore, thorough knowledge of the mission is helpful in selecting the test EME and determining appropriate procedures for exercising the ordnance system under test (SUT).

4.2.2 Ordnance Description. Complete identification of the item to be tested is one of the most important, yet often least-emphasized aspects of test preparation. Test results cannot be documented properly nor entered into DoD HERO databases without knowledge of all of the "identifiers." This may include identification at the EID, subsystem and platform/system levels, or some combination thereof; the Department of Defense identification code (DoDIC) or Navy ammunition logistic code (NALC), military nomenclature (for example, AGM-54, M2), part number (P/N), and national stock number (NSN). Reference documents, common names, and contract numbers are also helpful information. It is also important to identify all versions of the item for which the test data is applicable. Photographs and physical descriptions provide extremely useful documentation, especially if questions arise concerning which configurations actually were tested.

Associated with most configurations is a host platform/system from which the ordnance is carried or launched. The host platform/system, launcher, if required, and interface cables associated with a particular ordnance item can have a pronounced influence on the response of the EIDs. The ordnance when combined with its platform/system configuration is distinctly different from the isolated ordnance item, due to differences in the size, geometry, and interface

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of the constituent elements. This unique configuration can produce additional and more efficient coupling paths, especially at the lower frequencies where the increased size represents a more effective “antenna”. The platform/system, whether it is an aircraft, a shipboard launcher, an armored vehicle, or a person, often determines the first-order resonances of the overall configuration. Assuming that there is a path for energy to couple into the EID, the addition of a host platform/system can alter the distribution of currents in the vicinity of discontinuities on the ordnance enclosure, thereby altering the penetration into the enclosure. It is important that all designated platform/systems be included in the initial assessment. When only one platform/ system can be tested and multiple systems/platforms are used tactically, an analysis should be conducted to determine which platform/system is expected to cause the maximum EID response, and whether conclusions from that test can be applied to other platform/systems. Furthermore, an intra-system analysis or test should be conducted to determine if any of the platform transmitters can radiate the ordnance configurations so as to induce currents into EIDs that exceed the HERO criteria.

A general physical description should be provided that includes overall dimensions, weight, and materials used, for example, metal, composites, plastics, and so forth.

Part of the identification task involves assuring that the ordnance item to be HERO tested is an accurate representation of the item that will be fielded. HERO tests are normally conducted during Phase III (Production and Deployment) as defined in DoD Instruction 5000.85, "Major Capability Acquisition," on the first article or pre-production hardware (see 4.4.1.1.1).

4.2.3 Associated Equipment. Additional equipment, often referred to as ancillary equipment, may be necessary to support HERO testing and should be identified during the planning phase. Such equipment is often necessary to support handling and loading operations. It is important to consider all associated equipment in the planning and evaluation process and include such equipment since it is likely to influence EID response. Examples include shipping containers, handling equipment, such as cranes, carts, lifts, special tools, and diagnostic/ programming equipment. In addition, special test fixtures may be necessary to support the ordnance in a simulated post-launch configuration. (See 4.4.1.3 for additional guidance on the need for and use of ancillary equipment.)

4.2.4 S4 Description. When planning for a HERO test, it is essential to have a thorough knowledge of how the ordnance will be used throughout its service life or operational life cycle. The S4 is the progressive stages (phases) that begin at the time the ordnance is manufactured and continue until it is expended or reaches a safe distance from the launch vehicle/platform/system. This is depicted in FIGURE 1 below and defined in MIL-HDBK-240-4.

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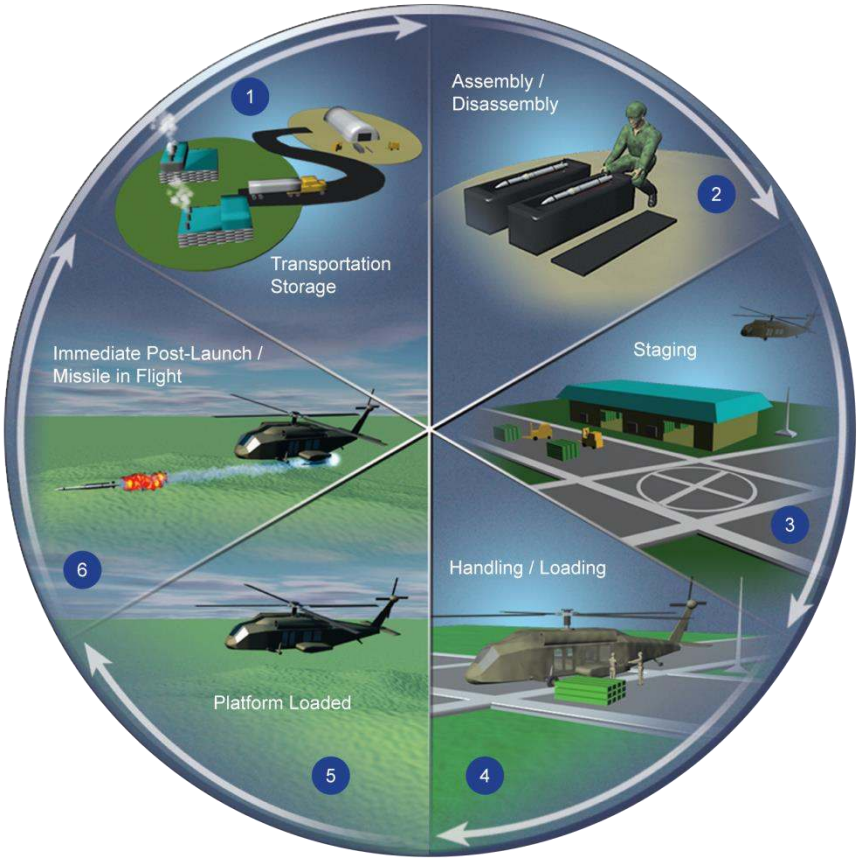


FIGURE 1. Stockpile-to-safe-separation sequence (S4)

An ordnance item may not always have six distinct S4 phases. For instance, when a missile leaves a factory it may be completely assembled prior to delivery to its deploying activity and, therefore, the assembly/disassembly S4 phase is not applicable to the activity. Furthermore, from a HERO standpoint, there may not be any significant differences between some phases. Therefore, testing just one phase may be sufficient to characterize the susceptibility of the ordnance system for other phases. In this case, each phase should still be addressed in the test plan and test report even if they are not planned to be tested, along with the justification. It is imperative to have a thorough understanding of each phase because this will define the three fundamental test parameters required to design the testing program and properly evaluate the ordnance system. The three parameters associated with each phase are the following: the physical configuration of the ordnance system and associated equipment, the specific operating procedures used to exercise the ordnance item, and the EME the ordnance is expected to encounter. These three test parameters should be assessed for possible HERO implications for each phase and determine the impact on test preparation to actual testing (see 4.4).

4.3 Pre-test Assessment. It is often possible to predict the effect, qualitatively, that certain design features will have on EID response to radio-frequency (RF) EMEs. In addition, the effect of certain ordnance handling procedures is often qualitatively predictable. Such predictions are valuable when planning which areas to emphasize or de-emphasize during the test. For example, it is important to identify potential points of entry (POEs) for electromagnetic

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(EM) energy. Physical dimensions may be indicative of resonant frequencies and operating procedures that can negate protective design features. Thorough test preparation includes identification of those design features, physical characteristics, and operating procedures that can be expected to influence EID response. Then the appropriate engineering decisions concerning the test planning with respect to test EME levels, configurations, and procedures can be made. Subsequent paragraphs provide further discussion of physical and electrical characteristics relevant in the HERO test planning process, as well as tools available for response prediction.

4.3.1 EID Description and Characteristics. Collection and analysis of EID information is critical to predicting potential susceptibilities and determining instrumentation requirements. In addition to proper EID identification (P/N, NALC, DoDIC, or NSN), the following information should be obtained and evaluated.

a. Type of EID.

b. Sub-component containing the EID.

c. Firing sensitivity.

d. Bridgewire resistance.

e. Response time.

f. Function within the ordnance.

g. Firing consequence (safety/reliability).

TABLE I is an example of the above information that should be part of the test plan and test report.

In addition, RF sensitivity data is useful, if available. A table similar to TABLE I should be developed for each EID contained within the ordnance. Subsequent paragraphs provide an explanation of the above parameters and their impact on test preparation.

TABLE I. EID Identification/Characteristics Example

Name/ Nomen-clature	DoDIC/ NALC or P/N	Type of EID	Ordnance Subsystem	Function / Application	MNFS (amps)	BW Resist. (ohms)	Function/ Response Time (msec)	Firing Conseq.
MK 1 MOD 0 Squib	M265	BW	Rocket Motor	Motor Ignition	0.200	2-5	20	Safety

4.3.1.1 Types of EIDs. There are many different types of EIDs used in ordnance. Examples include hot bridgewire (HBW), exploding bridgewire (EBW), semi-conductor bridge, carbon bridge, conductive composition, exploding foil initiator (EFI), and fusible link. The test engineers should understand the characteristics of the different types of EIDs in order to select

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the best-suited instrumentation and plan appropriately for the test. TABLE II provides a matrix of the most common EID types, along with some of the inherent characteristics that can impact their EM susceptibility. From the table, it can be seen that a suitable sensor is not presently available for the carbon-bridge or conductive composition EIDs. Consequently, evaluation of these devices relies on go/no-go techniques (see 4.7.5). When using temperature sensors with EBWs (non-gapped), EFIs, and fusible links, it will be necessary to establish the minimum fusing or detectable damage current in order to establish the required margins.

TABLE II. Typical EID Characteristics

EID Type	Typical Firing Sensitivity	Typical Response Time	Instrumentation Type
HBW	Moderate to High	1-100 msec	Temperature Sensor
EBW	Low	< 6 μsec	Temperature Sensor
Carbon Bridge	High	< 1 μsec	Not Available
Conductive Composition	High	≤ 300 μsec	Not Available
EFI	Low	< 2 μsec	Temperature Sensor
Fusible Link	Low to Moderate	1-100 sec*	Temperature Sensor

* sec = second

EBWs and EFIs require a significant amount of energy (high voltage) to initiate. Through analysis and testing, it has been determined that the typical DoD EME (peak or average) produced by current operational radar and communication systems cannot initiate these devices through direct coupling. However, it has been demonstrated that these devices may be damaged by the power dissipated in the bridge or foil to the point where they may not initiate when the proper firing signal is applied. Therefore, these devices should be evaluated for performance degradation (reliability) using traditional temperature-sensing techniques. Typically, an EFI is used as the initiating device in Electronic Safe and Arm Devices (ESADs). The electronic circuitry in the ESAD provides both triggering and high-voltage switching functions. Both of these functions are potentially upset by the EME. Therefore, conducting HERO testing with the circuit powered to detect upset of the ESAD electronics may be necessary. Any upset to the ESAD electronics will be evident by the ESAD applying high voltage to the firing capacitor or high voltage being discharged through the EFI. Any HERO testing that solely measures the voltage across the firing capacitor does not assess potential ignition of an EFI. In most EFI systems, two electrical inputs are required: a high voltage input across the capacitor and a trigger electrical/logic input. In the absence of either of these two inputs, no electrical firing stimulus can be applied to the EFI. Some EFI systems do not charge and wait for a trigger stimulus. After the trigger logic has been satisfied, the capacitor is charged and the EFI is initiated as the capacitor voltage reaches the EFI firing circuit breakdown voltage. Each EFI system's operation must be analyzed for each system's application to determine when the EFI system would be deemed armed. An EFI system will be considered to be armed when the EFI system is awaiting the final stimulus required to energize the EFI. EBWs and EFIs are generally considered to be insensitive EIDs since they can only be fired via a high-voltage firing pulse that is provided from the discharge of a capacitor. However, if the EFI system is armed during any HERO phase prior to safe separation, the system will be required to perform the modified EME operational testing for each applicable phase in its respective armed state. This represents a safety-critical function

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that MIL-STD-464 states must demonstrate a margin of 6 dB. To demonstrate this margin, the peak EME levels to which the system is exposed during testing when the firing circuit is in an armed state can be defined by the system's operational EME with a safety margin of 6 dB, but not to exceed the peak EME levels for HERO defined in the Maximum External EME Levels for Ordnance table in MIL-STD-464. The operational EME for the system may be defined by the EMEs in the external RF EME tables of MIL-STD-464 and/or MIL-HDBK-235 or the system specification.

4.3.1.2 Firing Sensitivity. Typical firing sensitivity is a determination of the threshold initiation point (relative to other device types) when the EID is subjected to electrical stimuli, either voltage or current, and is described in terms of the maximum no-fire stimulus (MNFS). High sensitivity implies that the device is more sensitive to applied signals than most other devices, while low sensitivity implies that the device is relatively insensitive. The MNFS is determined using one of the following methods:

4.3.1.2.1 One-amp/one-watt rating – The EID should not fire within 5 minutes when subjected to a current of 1 amp minimum per bridge with an associated power of 1 watt minimum per bridge. This applies to each wire of devices having more than one bridgewire.

4.3.1.2.2 MNFS – The greatest firing stimulus that does not cause initiation within 5 minutes of more than 0.1 percent of all EIDs of a given design at a confidence level of 95 percent. When determining the MNFS for EIDs with a delay element or with a response time of more than 5 minutes, the firing stimulus should be applied for the time normally necessary for actuation. There are a number of statistical test methods that may be used to determine the EID's MNFS according to the definition above, such as the Bruceton, Probit, and Langlie.

The EID manufacturer usually determines the firing sensitivity; however, other commercial companies and military activities are capable of performing the required assessments. It is essential that firing sensitivity data and the method used to determine the MNFS be obtained and analyzed during the planning process.

4.3.1.3 Bridgewire Resistance. EID bridgewire resistance is required for a HERO evaluation. Typically, the resistance is given as a range; for example, 4-8 ohms or 1 ± 0.1 ohm. The variation of the resistance for a given EID can have a significant impact on the EID and should be considered, especially if there is a wide range of resistance.

4.3.1.4 Response Time, τ. It is important that the EID response time, or time constant, τ, be determined and analyzed with respect to the modulation characteristics of the emitters expected to be encountered by the ordnance and the transmitters used to generate the test EME. This is especially important for EIDs that are very fast responding devices (for example, microseconds (μsec)) that could respond to peak field strength levels. Notable examples of fast responding EIDs are conductive mix devices, such as 20-millimeter (mm) electric primers.

The response of bridgewire EIDs to a current step input is characterized by an exponential rise in temperature. Response time is defined as the time it takes for the bridgewire temperature (or current) to reach 63 percent of the final (equilibrium) value. (An alternative measure of the time

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response of an EID is “rise time,” the time it takes to change from 10 percent to 90 percent of the final temperature.)

The time that elapses from the application of the firing stimulus until the EID completes its intended function is referred to as the function time of the EID. Function times include both the time to reach the critical firing temperature and the time required for the energetic material to respond to the applied energy.

Typical time constants for bridgewire devices are between 1 and 20 milliseconds (msec). Heating and cooling time constants are similar. EIDs with thermal response times less than or equal to the radar pulse width are referred to as “pulse-sensitive” or “peak power-sensitive” devices. Examples include conducting composition devices, thin film devices, and semiconductor junction devices. As a general rule, tests should be performed using both peak and average field strengths for “pulse-sensitive” or “peak power-sensitive” devices.

4.3.1.5 Thermal Stacking. “Thermal stacking” can occur when multiple radar pulses induce current within the EID’s response time. In such cases, the bridgewire temperature can incrementally increase (“stack”) with each radar pulse because there is insufficient cooling time between pulses. FIGURE 2 illustrates this phenomenon.

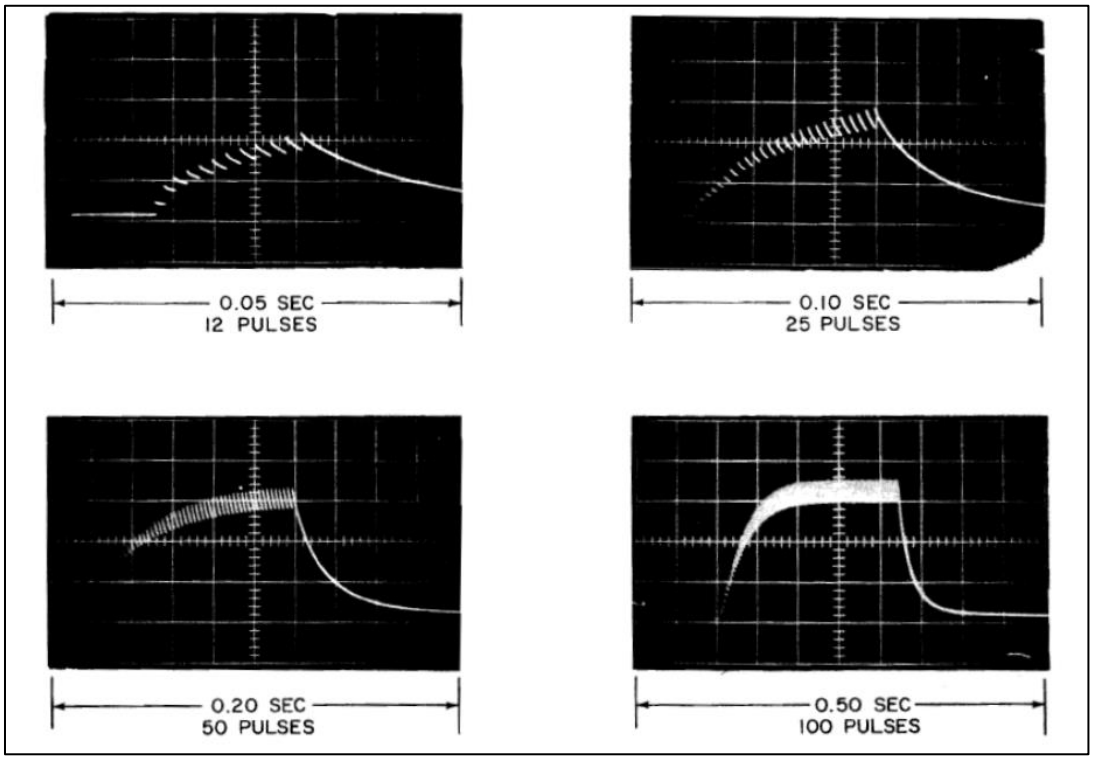


FIGURE 2. Illustration of thermal stacking of MK 1 squib during 9 GHz radar test

It is recommended that EID time constants be determined as standard practice. When the thermal time constant of an EID is known, calculations can be made to assess responses for varying emitter parameters. If the response of an EID is known for continuous wave (CW), then

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a meaningful response figure for a particular pulsed emitter can be obtained by using the following multiplying factor (MF) for peak power in the pulse:

$$MF = \frac{\left(1-e^{-t_2/\tau}\right)}{\left(1-e^{-t_1/\tau}\right)} \quad (4\text{-}1)$$

where:

t_1 = radar pulse width
t_2 = radar pulse interval = 1/PRF (pulse repetition frequency)
τ = minimum EID time constant

For example, assume an EID has a time constant of 100 μsec and a maximum no-fire power (MNFP) of 1 watt for CW illumination. For a radar system with a pulse width of 30 μsec and a pulse interval of 1000 μsec, the MF is:

$$MF = \frac{\left(1-e^{-1000/100}\right)}{\left(1-e^{-30/100}\right)} = 3.86 \quad (4\text{-}2)$$

Therefore, the maximum no-fire (MNF) level for the EID for the radar in the example is 3.86 watts. Similarly, the MF can be used with known responses from radiated fields. If the installed EID is capable of tolerating 100 milliwatt per centimeter squared (mW/cm^2) for a CW field, then it is reasonable to assume it can tolerate 386 mW/cm^2 peak power density for the particular radar. Similar calculations can be made to compare peak electric fields (E-fields), voltages, and currents to CW parameters; however, the square root of MF should be used to obtain correct values. If a margin of 16.5 dB exists for the CW field, then the same margin of 16.5 dB exists for the calculated pulsed field.

When the EID time constant is short compared to both the emitter pulse width and pulse interval, the MF approaches one, as expected, indicating that a single emitter pulse has the same effect as CW.

4.3.1.6 EID Function. EIDs may be used to perform a variety of functions within the ordnance. Some EIDs can perform more than one function; therefore, it is important to identify the function within the ordnance item/subcomponent. Examples of EID functions are rocket motor igniters, warhead detonators, explosive switches, cartridge-actuated devices, battery initiators, and cable cutters. In addition, the EID function will have a significant impact on its firing consequence. EIDs of the same type and make may be used in different sections of the ordnance. It is important that the subcomponent (for example, fuze, battery, igniter, or gyro) containing the EID be identified and recorded in the test plan (see paragraph 5).

4.3.1.7 Firing Consequence. The actuation of an EID for a given function will have a resulting consequence. For DoD applications, the firing consequence has been divided into two categories: those that will result in system performance degradation (reliability consequence); and those that will result in a hazard to personnel, damage to the system, ancillary equipment, facilities, platform/systems, and so forth, or disable safety features that may lead to a catastrophic event (safety consequence).

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During the planning process, it is important that an analysis be performed on each EID of a given application to determine the consequence of inadvertent firing. The analysis should be performed and documented by qualified, explosive-safety personnel. In addition, when examining EID functions and firing consequences, it is essential that the associated firing circuits also be examined to fully understand the operation of the system and the impact on the firing consequence.

Safe and Arm (S&A) circuits play an important role in determining the firing consequence (safety and reliability) of an EID contained in an ordnance system. Therefore, it is essential that the test preparation process include a description and understanding of the S&A concept incorporated into the ordnance system. MIL-STD-1316 requires that S&A devices be designed with safety features that, in most instances, preclude the existence of a safety consequence. One example of a safety consequence is the actuation of an initiator used in an S&A device with an in-line explosive train. However, initiators used in these designs require relatively high voltages to fire these devices, so they should not fire at stimuli below 500 volts. Therefore, direct initiation by EM energy is highly unlikely. However, there is a potential for inadvertent initiation of these devices due to EM energy upset of the trigger circuit. In some instances, an EID that normally is used to remove a safety feature in the immediate post-launch environment will cause the malfunction of the S&A device in the ordnance item if it is initiated prematurely.

4.3.2 Evaluation of RF-Protective Features and Weaknesses. The following paragraphs discuss the design aspects of ordnance that should be considered when defining elements of the HERO test program, including test points. Some ordnance items have extensive protection incorporated into the design, whereas others suffer from an absence of protective features. Notable protective features or the lack thereof, influence the test plan and, ultimately, the testing process.

There are numerous design guides and handbooks that provide much more detailed information on EM protection techniques, as well as engineering data useful in estimating resonant frequencies and identifying potential POEs. These include NAVAIR AD 1115, MIL-STD-331, and NAVSEA OD 30393.

4.3.3 Enclosures. Enclosure shielding is one of the most important features for the protection of ordnance against incident energy. The test engineer should be aware of weaknesses in the enclosure shielding integrity. Enclosure discontinuities such as seams, ports, vents, access panels, rocket motor nozzles, seeker windows, or use of composite materials may reduce the overall shielding effectiveness (SE) of the enclosure. A good understanding of the shielding design of the ordnance helps predict potential areas of susceptibility; for example, POEs, or resonant frequencies. It is important to identify types of materials and their specified and actual SE.

4.3.3.1 Cables and Connectors. Ordnance typically contain both power and signal cables that can act as antennas, particularly when exposed to high frequency (HF), very-high frequency (VHF), and ultra-high frequency (UHF) EME levels. The cable shielding design should be assessed to identify potential strengths and weaknesses. The assessment should take into consideration the length of the cable, type of shield, type of connector, and connector or shield termination. Knowledge of the cable shielding design would help predict whether

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potential susceptibility problems may occur. In addition, the type of cable termination is extremely important because the effectiveness of shielded cables can be compromised if the shield is not peripherally terminated at the connector. It is noted that direct current (DC) bonding measurements are not an adequate means of determining the SE of the connector/shield termination. MIL-STD-1377 can be used as a guide for testing suspected weaknesses in cable and connector shielding design.

4.3.3.2 Electromagnetic Interference (EMI) Gaskets. Ordnance may contain gaskets where there are shield discontinuities. The gasket SE should be assessed with respect to material properties, such as conductivity, compressibility, surface contact area, corrosion, and so forth, as well as installation techniques, compression force, and material properties of the mating surfaces where the gasket will be installed (surface area, conductivity, and surface preparation (paint, chemical coating)).

4.3.3.3 EMI Filters. EMI filters can be incorporated to protect signal and power lines. The test engineer should determine the EMI filter type, either absorptive or reflective, and the electrical characteristics, such as frequency attenuation level and power rating. Such characteristics may indicate a potential for resonance (reflective-type filters), or where protection is minimal; that is, frequencies below which no attenuation is provided. Filtering data and the method used to derive the attenuation values should be obtained for analysis. It is noted that insertion loss measurements of individual filters using current test methodologies do not always apply once the filter has been installed in its actual application. In addition, improper installation techniques can also compromise filter effectiveness. RF arcs contain components at all frequencies. Since an EMI filter is designed to preclude only high frequencies from the EME and pass low-frequency firing signals, it is not capable of discriminating between the components of an arc and the intended firing signals. Therefore, special techniques should be employed to provide protection against the HERO problem caused by arcing. In general, there are two methods that are used:

a. Provide open contacts in the firing system between the filter and the EID, or

b. Reduce the RF potential of the mating power contacts to zero prior to the final connection of the firing circuit to the ordnance item.

If either or both of these techniques are followed, the HERO test engineer will have some reasonable assurance that RF arcing will not be a problem.

4.3.4 Firing Systems. For the purposes of this handbook, a firing system consists of a power source, transmission line, and all switching circuits that are required to transfer or control firing energy to an EID (FIGURE 3). Because the firing system provides the path for transferring firing energy to the EID, it also can provide a path for transferring EM energy to the EID. Each element of the firing system's design should be assessed for possible weak links with respect to the overall protection scheme. This is especially true for firing circuits that require an electrical connection external to the ordnance enclosure, as is generally the case.

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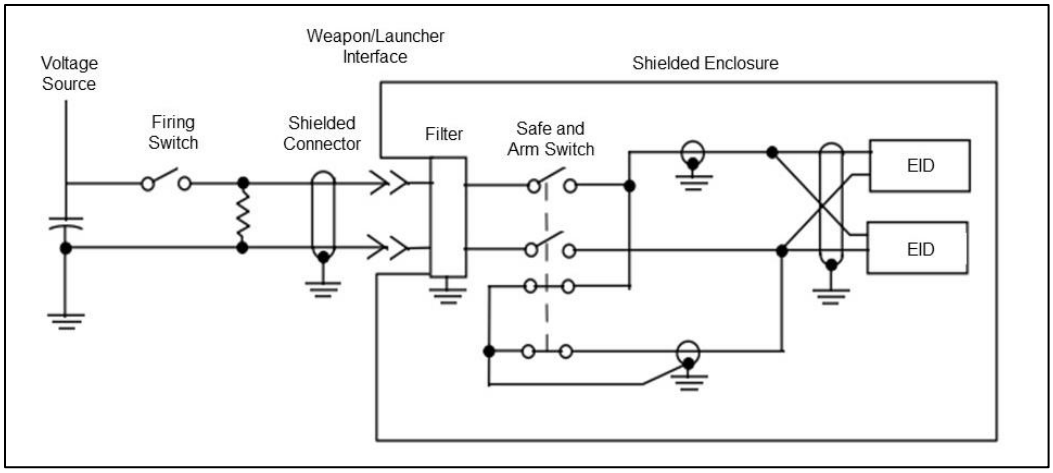


FIGURE 3. Typical HBW firing circuit with an S&A device

4.3.4.1 Typical EID Circuit Design. There are two modes of unwanted excitation in an EID or its firing circuit, the differential mode and the coaxial mode. In addition, the unwanted signal may be introduced into the ordnance and propagate to the EID through the same path designed for the firing circuit, or it may propagate to the EID through capacitive or inductive coupling via an undesired path, such as a ground loop.

Differential mode RF excitation occurs in two-wire firing circuits. EM energy propagates to the EID or its firing circuit between two wires in the same manner as the normal alternating current (AC) or DC firing current. This will cause joule (resistance) heating of the bridge material, thereby causing inadvertent initiation or dudding of the EID. FIGURE 4 illustrates the differential mode of excitation. In this mode, it might appear that if a large mismatch of impedance occurs between the EID or its firing circuit input and the transmission line, which is usually the case, most of the EM energy would be reflected at the EID or firing circuit input. Although most of the energy is reflected, enough can be coupled to the EID to cause the bridgewire to heat.

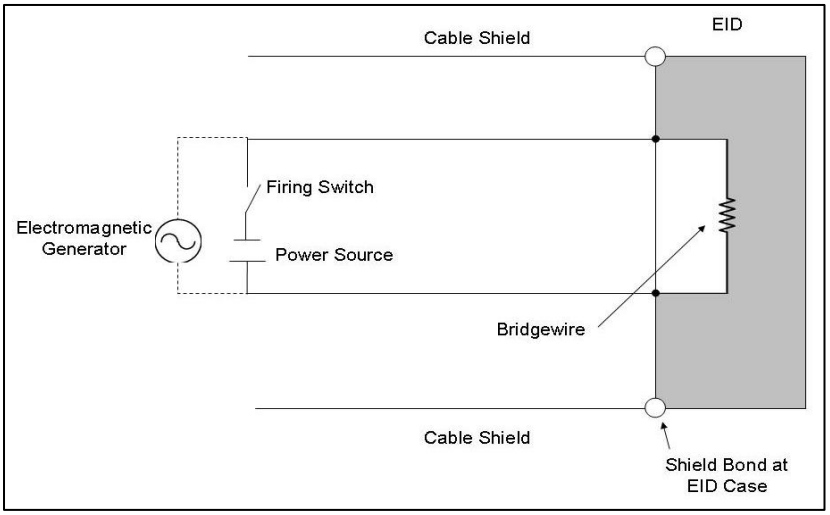


FIGURE 4. Differential mode of RF excitation in a two-wire firing system

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In a coaxial firing system, the EM energy propagates to the EID or its firing circuit along one or both legs of the normal AC or DC firing paths. The unwanted signal circuit is completed via a ground loop or capacitive/inductive coupling at the EID. This can be visualized easily by considering a wire or metal rod center conductor contained inside a cylindrical conductor, such as a shield with which it is concentric (FIGURE 5). The EID bridge material is connected at each end to the center and outer conductors. The unwanted signal voltage appears across the EID bridge as a result of an undesired path that occurs between the center and outer conductors, either through a ground loop or a capacitive/inductive coupling. Here again, heat in the bridgewire material is generated by EM energy just as is the intended AC or DC firing current.

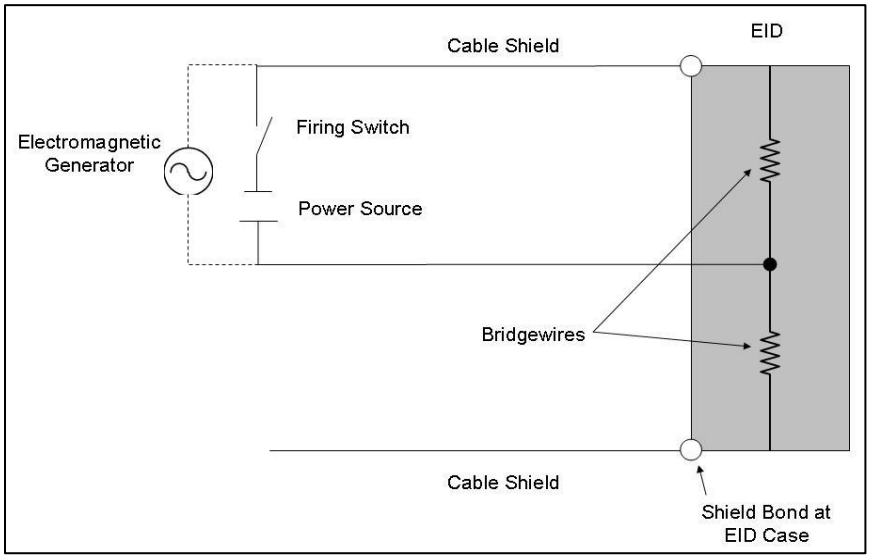


FIGURE 5. Coaxial mode RF excitation in a coaxial firing system

The coaxial mode of RF excitation also can occur when using a two-wire, balanced shielded system through any high-impedance connection in the shield continuity. In this case, the two lead wires serve as the center conductor and the shield serves as the outer one (FIGURE 6). In a two-wire balanced system, energy transferred to the EID in the coaxial mode will cause a high potential to be developed from the bridge through the explosive mix to the EID case. This can cause RF arcs to occur in the explosive mix or can cause dielectric heating of the explosive mix.

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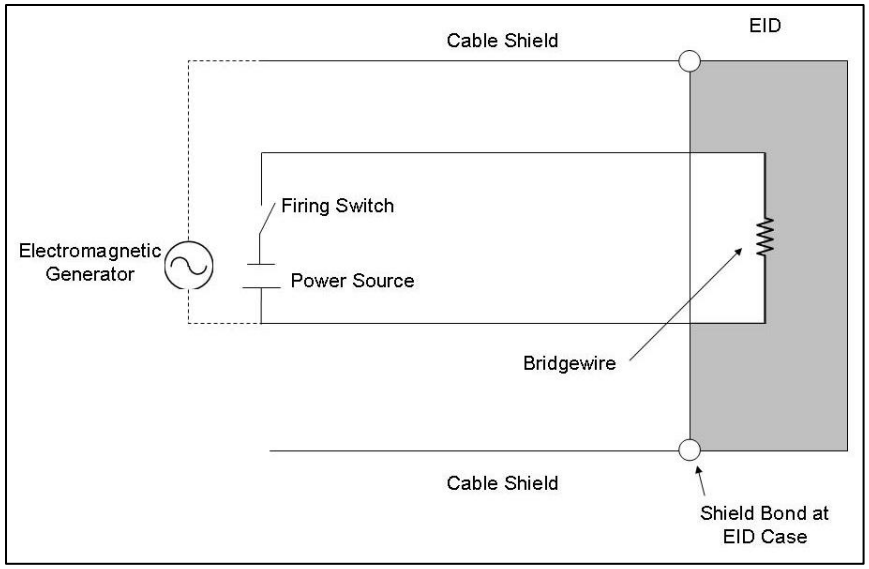


FIGURE 6. Coaxial mode RF excitation in a two-wire firing system

4.3.4.2 Energized vs. Non-Energized Firing Circuits. HERO susceptibility thresholds, or those minimum EME levels where susceptibility occurs, can vary, depending on whether the ordnance item is powered or not powered during test. Energized firing circuits may impact the coupling of energy to the EIDs by altering the electrical path by opening/closing circuits. In addition, EM energy can be coupled easily into components, relays, micro-switches, or powered firing circuits, causing energy to be applied to the EID. The test preparation process requires that the ordnance be analyzed to determine those S4 phases in which power may be applied to the EID firing circuits. This process has a significant impact on the selection of the test EME, test procedures, ordnance configuration, and additional test assets. For example, if the firing circuits are energized, both peak and average test EME levels are required to address both the fast response times of electronic firing circuits and the comparatively slow response times of the EIDs themselves (see 4.5.1.2 for guidance on selection of EME levels, peak and average, for testing energized and non-energized circuits.)

4.3.4.3 In-Line vs. Out-of-Line Circuits/Explosive Trains.

4.3.4.3.1 In-Line Circuits/Explosive Trains. An in-line EID is so called because it is always aligned with the explosive train that it is intended to initiate. Consequently, if the EID should be inadvertently initiated by EM energy, the device in which it is employed will function as intended. In some applications, such as the operation of gas generators or explosive bolts, the EID may be initiated automatically upon receipt of signals from timers or sensors that respond to stimuli experienced during the fire cycle of the ordnance system. The reliability, EM vulnerability, and any potentially hazardous failure modes of the firing circuit associated timers and sensors need to be examined carefully during the test planning process.

In addition, two types of electronic arming devices have been developed for military use, the ESAD and the electronic safe, arm and fire device. These devices utilize an in-line firing circuit and explosive train. Both devices have sophisticated firing schemes that usually contain a capacitor that can be charged to 2 kilovolts (kV) to provide a relatively high-voltage firing pulse

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to initiate the EID. Also, these devices use very insensitive EIDs, such as EBWs and EFIs, to initiate the explosive train.

It is noted that there is a potential for RF affecting the in-line circuits/explosive trains if the system is not properly shielded. RF can upset the trigger circuit, causing the high-voltage firing pulse to initiate the EID. Monitoring these circuits for RF-induced firing stimuli is challenging and often resorts to go/no-go techniques.

4.3.4.3.2 Out-of-Line Circuits/Explosive Trains. An out-of-line EID is so called when, until armed, the EID explosive output is either misaligned with the secondary explosive train or a mechanical shutter interrupts the explosive train. Also, the EID firing circuit may contain EID-driven switches or rotors that interrupt the continuous electrical path to the power source. Mechanical shuttered systems normally are employed in conjunction with low-energy devices. Typically, this design is used in warhead fuzes or rocket motor arm/fire devices. These devices should be analyzed thoroughly to understand the arming sequence and determine the impact of the susceptibility of the EIDs.

4.3.5 Multiple Platform/System Analysis. For ordnance used on multiple platforms/systems, be aware that the dominant resonant frequencies are likely to be platform/ system-dependent, especially at HF and VHF frequencies. An analysis should be undertaken to determine the “worst-case” platform/ system configurations and frequencies that should be tested. The appropriate platform(s) is (are) selected based on test objectives and practical constraints of the test. Caution should be exercised when extending test results to platforms other than the one used to conduct the test. All platforms should be addressed in the test plan with justification for any platforms that are not considered “worst-case” (and therefore not tested).

4.3.6 Response Prediction. There is value in predicting the conditions expected to stimulate the greatest EID responses.

These conditions include specific EME parameters, such as frequency, illumination angle, and polarization, the various physical configurations of the ordnance/associated equipment, and the procedures used to exercise the ordnance. Predictions are especially helpful for focusing the test where maximum responses might be expected. This is especially important when time or equipment constraints preclude testing all possible combinations of the aforementioned conditions. Predictions also are useful in prioritizing the generation of certain of these conditions, to increase the likelihood of detecting the greatest responses. There are several tools worthy of consideration. These tools are discussed in the following paragraphs.

4.3.6.1 Results of Previous Tests. Perhaps the most useful prediction tool is simply the results of previous HERO tests on similar items. While there is no guarantee that the results will repeat themselves on the current item, trends may be evident that may suggest conditions that are not likely to cause significant responses. If previous evidence shows that portions of the test can be verified through analysis, then the information should be included in the test plan and approved by the Service technical authority. It is always advisable to invest some time into researching whether test data exists on similar systems.

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4.3.6.2 Analytical Prediction of Resonances. Predicting the frequencies at which EID responses exceed specified allowable levels is challenging. The purely analytical approach is complicated by the fact that the magnitude of RF currents that couple into EIDs can be influenced by the characteristics of both the outer enclosure and the internal EID cables. At microwave frequencies, ordnance enclosures are relatively large compared to the wavelength, resulting in complex internal standing-wave patterns. These field patterns are complicated further by the complex, irregular internal geometry of actual systems. It is difficult to predict the magnitude of RF currents that couple into EID cables within such environments. Finally, since the impedance mismatch between the EID and the attached cables is generally unknown, it is impossible to determine the absolute magnitudes of EID currents. However, it is relatively easy to calculate resonant frequencies associated with overall physical dimensions of the weapon/host platform/system, cables, and apertures. If the uncertainty of the interactions within the weapon enclosure is ignored, the simple resonances attributed with the physical dimensions mentioned above are reasonable candidate frequencies that may warrant investigation.

4.3.6.3 Analytical Tools. There are various software tools that can be used to predict the response of EIDs when the ordnance item is exposed to defined EME levels. The MAE can be calculated if the user specifies limits on EID currents. An MAE analysis tool has been developed that predicts the highest field strength levels to which bridgewire-type EIDs can be exposed without exceeding established limits for EID response, and depicts the MAEs in both graphical and tabular formats. For access to this tool, contact your technical point of contact (POC) listed in 11.6. Although the program primarily was intended for predicting MAEs in the absence of testing, it can be used as a pretest aid. The program provides a conservative estimate of the most susceptible frequency and the associated MAE level for a particular ordnance configuration. It utilizes a matched, half-wave dipole model to calculate the response of the EID of interest. EID MNFP is determined from the current sensitivity maximum no-fire current (MNFC) and bridgewire resistance. Then an EID response limit is calculated on the basis of a selected firing consequence. The program is based on a combination of simple dipole theory and empirical data. For test planning, the output of the MAE Analysis Program is useful in three ways. First, the program predicts minimum MAEs for user-defined ordnance and ordnance-platform/system configurations. Frequencies where MAE levels are minimal are candidates for special attention, as are predicted resonant frequencies. Second, one might consider reducing the number of test frequencies in bands where predicted EMEs are less than the performance requirement MAE levels. Lastly, at frequencies where predicted MAEs are extremely low, such as less than 10 volts per meter (V/m), the test engineer may elect to begin evaluations at levels below the full requirement or reduce instrumentation sensitivity to prevent overstressing and risking burnout of the EID instrumentation.

4.3.6.4 Reverberation Chamber. In addition to the identified prediction methods there is value in quickly accomplished broad frequency range measurements to screen for EID responses. One measurement technique that is most useful uses a reverberation chamber to quickly scan for EID responses at shorter wavelength frequencies (generally above 1 gigahertz (GHz)).

A reverberation chamber is an RF tight enclosure, which is excited by an RF source at frequencies where the chamber dimensions are large compared to wavelength. The simultaneously existing modes are "stirred" by a field-perturbing device referred to as a “paddle wheel tuner”. As the tuner assembly is rotated by means of a stepper motor mounted in the

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chamber, the complex standing-wave patterns are significantly altered and the SUT is exposed to randomized fields from all directions and polarizations. In a properly operated reverberation chamber, the EME is isotropic and randomly polarized. These characteristics provide an all aspect angle illumination of the SUT.

Reverberation chamber screening of a particular SUT configuration can be accomplished quickly over a broad frequency range as compared to the broadly spaced discrete frequency, manually pointed antenna method commonly employed at an Open Area Test Site (OATS). The efficiencies gained by illuminating the SUT at all aspect angles along with electronic tuning increases the speed at which testing can be accomplished.

Screening methods employing reverberant techniques identify frequencies and fields that result in EID responses in the SUT. Greater concentration on these frequencies and fields can then be given during the conduct of the OATS test.

4.3.6.5 Ordnance Operating Frequencies. When evaluating energized firing circuits, the operating frequencies of the ordnance subsystems should be considered. Caution should be exercised when testing in-band receiver frequencies to avoid burnout of front-end circuits. It is advisable to begin testing receivers at low power levels and gradually increase them to high levels or take steps to prevent in-band energy from damaging front-end circuits.

4.3.6.6 Supplemental Tests. The use of additional test sites and techniques to identify system responses or safety margin can be beneficial when combined with traditional HERO testing. Reverberation or mode-stirred chambers are a highly effective means of performing swept-frequency testing to identify EID responses for additional frequencies beyond standard test protocol.

Typically, frequencies above 1 GHz can be changed quickly and swept to uncover resonances that would otherwise not be detected in discrete frequency testing. These frequencies and fields can be added during the conduct of the OATS test to ensure better results. System-level testing at the full external threat environment has historically been the primary method employed when conducting external RF EME evaluations. However, due to increasing EMEs this may not always be feasible. The use of tools and techniques like the electromagnetic probability-of-effect assessment tool (EMPAT) can be used in conjunction with traditional HERO testing to assess the safety margin of systems in an EME. In situations where test methods cannot meet the higher power level requirements because existing test equipment and facilities cannot generate the necessary RF power, EMPAT provides a method to assess the system given the item meets certain test criteria. The supplemental method is to decompose the testing into multiple lower power tests and then statistically determine the probability-of-effect. The distribution of the maximum fields within the system enclosure is first measured using complex cavity test techniques. This can be scaled to the external threat environment by the frequency-dependent SE transfer function of the system enclosure. Then, the internal equipment threshold can be directly measured through susceptibility testing (at levels representative of internal fields). Finally, the probability-of-effect is determined from the system hardness margin (for example, the ratio of equipment threshold to the maximum environment). It should be noted that these tests are not substitutes for HERO ground plane testing, but are accepted supplemental tests when used in addition with traditional OATS testing.

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4.4 Test Configuration. As stated in 4.2.4, it is extremely important to identify the three fundamental parameters associated with each phase of the S4, because they will ultimately drive the test requirements. Again, the three parameters associated with each phase are the following:

a. Physical configuration of the ordnance item and associated equipment.

b. Specific operating procedures used to exercise the ordnance item.

c. The EME the ordnance item is expected to encounter.

These three parameters should be assessed for possible HERO implications for each phase, and the impact on test preparation and actual testing should be determined. Each of these parameters is discussed in more detail in the following paragraphs.

4.4.1 Physical Configurations. It is important to recognize that the ordnance item may be configured differently as it transitions through the various phases of its S4. Each configuration may, therefore, exhibit significantly different responses to the EME; that is, it may be very susceptible in one configuration, but completely unresponsive at another. Test planning requires a decision regarding which configurations should be tested. This decision is based on a pretest assessment of the protective features and weaknesses, which may be influenced by data from previous HERO tests on similar items. In addition, different versions of the ordnance may have their own unique physical configurations for any or all of the phases. Therefore, identify and analyze all versions of the ordnance (that is, tactical, training, telemetry, or dummy). A comparative analysis should be performed on all versions to determine which versions should be tested.

Physical configurations are defined in the following paragraphs to include the specific configuration of the ordnance item, its launcher, host platform/system, and interface cables, as well as any associated equipment and personnel needed to support the testing. An analysis should be conducted to define what constitutes an acceptable representation of the physical configurations for each phase of the S4.

4.4.1.1 Ordnance SUT. The selection of the ordnance configuration to be tested is a very important part of the planning process. Consider the transitional configurations of the ordnance item for the various phases of the S4, and issues related to the following:

a. Pre-production/first article versus production configuration.

b. Explosively loaded units versus inert units.

c. Tactical configurations versus training and telemetry versions.

d. Multiple host platform/systems, launchers, interface cables, diagnostic equipment, and so forth.

4.4.1.1.1 Pre-Production/First Article vs. Production Item. Testing of prototype hardware early in the acquisition cycle may be necessary from a programmatic standpoint.

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However, HERO testing on pre-production hardware, normally considered "developmental tests," usually is reserved for extremely complex systems where delays cannot be tolerated due to deployment deadlines. Once the production item is available, the Service Technical Agent will review the changes. If there are no design or manufacturing changes with respect to HERO, the two versions may be deemed similar. Otherwise, the Service Technical Agent will notify the Program Office and recommend further testing.

4.4.1.1.2 Explosively Loaded vs. Inert Test Articles. HERO tests almost always are conducted using inert, production-representative units; that is, all explosives, pyrotechnics, propellants, fuels, and chemicals have been removed or replaced with inert materials. On rare occasions, HERO testing may be performed using ordnance items with explosively loaded EIDs. If explosively loaded EIDs are required for HERO testing, then standard operating procedures (SOPs) for explosives should be strictly followed. These SOPs will most likely be unique to each Service or test facility (see 4.7.5 for non-instrumented testing).

4.4.1.1.3 Tactical Configurations vs. Training and Telemetry Versions. A number of ordnance items, especially, missiles, rockets, and torpedoes, will have training or telemetry versions. A HERO assessment should be performed comparing the training/telemetry versions to the tactical configuration to determine the differences with respect to their EM susceptibility characteristics. If significant differences are noted, then those versions that are different with respect to their RF susceptibility characteristics will also have to be HERO tested.

4.4.1.1.4 Multiple Host Platforms, Launchers, Interface Cables, and Diagnostic Equipment. Ordnance may be authorized for use with multiple platform/systems, launchers, interface cables, and diagnostic equipment. All versions of these systems and equipment should be considered for HERO because they can have significant impact on the susceptibility of the ordnance item. An example of multiple platform/systems would be one missile that can be launched from a tank, ground vehicle, helicopter, or human.

4.4.1.2 Ordnance Configurations. The following paragraphs provide discussions on the selection of the ordnance configurations for the various phases of the S4.

4.4.1.2.1 Transportation/Storage. It is important to determine the configuration of the ordnance system when it is being transported or stored. Some systems are shipped as all-up rounds (AURs), meaning they are completely assembled, while others are shipped as sections or components. It is obvious that AURs containing EIDs should be considered for HERO evaluation. However, it is not so obvious that system components containing EIDs be evaluated for HERO. Components containing EIDs that are shipped and stored in individual containers should be considered for evaluation, especially if they are packaged in non-metal containers. Components may be extremely susceptible to EME levels since their internal electronics/wiring/EIDs may be directly exposed to the EME. Evaluation of system components/sections can be very difficult and complex to determine at the time of a HERO evaluation because at this stage of the acquisition process the concept of operations, packaging, and end users may not be finalized by the program office. Nonetheless, these item configurations should be identified and evaluated as part of the overall test program utilizing the best available information on proposed configurations at the time.

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4.4.1.2.2 Assembly/Disassembly. Ordnance items shipped as sections or components will require assembly/disassembly sometime during their deployment cycle. This usually occurs at the field activity or onboard a Navy ship. The assembly/disassembly process should be reviewed to determine the components involved in the procedures to be performed, the location where the components will be assembled or disassembled, and the expected EME at this location. If the expected EME at the proposed assembly or disassembly location exceeds the approved levels for these operations or it is not a controlled environment area, then the components and procedures for assembly or disassembly should be included as part of the testing effort.

4.4.1.2.3 Handling/Loading. The handling/loading S4 phase is one of the most critical parts of the HERO evaluation process since significant RF energy may be coupled into ordnance as a result of physical contact and/or electrical connections that are made between the ordnance and any metal structure of different electrical potential such as the launcher or platform. Personnel contact with the ordnance item, making and breaking electrical connections, performing diagnostic tests, and attaching or removing the ordnance item from large metal structures, such as launchers and platforms, have a pronounced impact on the EM susceptibility of the ordnance system. All configurations should be evaluated (analyzed, tested, or both) to ensure that sufficient test data is collected to assess all handling/loading effects.

Historical data has shown that these effects are prevalent during exposure to EMEs below 1 GHz where larger structures are more likely to exhibit resonance, causing an increase in the electrical potential of the launcher and/or platform. However, frequencies above 1 GHz have demonstrated little impact during performance of handling/loading procedures due to higher surface impedances of materials, which limits the propagation of electrical current and does not lead to any significant increase in electrical potential. Therefore, HERO testing during handling/loading procedures are often not performed at frequencies above 1 GHz.

While most ordnance items are unlikely to exhibit susceptibility during performance of handling/loading procedures above 1 GHz, there are circumstances where this may not be an accurate assessment and further evaluation is warranted. These circumstances include but may not be limited to impulse cartridges, which have exposed firing pins, and any ordnance item that involves some level of assembly during performance of handling/loading procedures such that the EM shielding afforded to internal EIDs and firing circuits is less than that provided in its platform-loaded configuration. Should it be determined that performing handling/loading HERO test procedures above 1 GHz are not required then the following statement should also be included in the test plan and report:

> "An analysis was performed on the ordnance system and determined the system will not be susceptible to the restricted EME levels during handling/loading procedures in an EME above 1 GHz; therefore, maximum allowable environments will be established at the restricted levels associated with the handling and loading phase for frequencies above 1 GHz."

4.4.1.2.4 Staged. Once the ordnance item has been prepared for loading operations, it is sometimes pre-positioned in a designated area until the loading operations actually occur. Usually, the designated area is not a controlled environment and, therefore, it may be exposed to

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high-level EME levels. In addition, the configuration of the ordnance item in the staged phase is often very similar to that of the transportation/storage phase (especially when those systems are shipped as AURs in non-metal or inadequately shielded containers) and the two phases may be evaluated simultaneously.

4.4.1.2.5 Platform-Loaded. The physical configuration of the ordnance item can be somewhat complex for this phase. The configuration will include the ordnance item, the launcher, the platform/system, and all interface cables, lanyards, and so forth. The firing circuits and other input cables in the test platform/system should be intact and operational. In addition, systems that have such devices such as safety pins and arm/de-arm mechanical switches that affect the EM energy coupling to EID firing circuits should be evaluated in all appropriate positions. The configuration and interface of a launcher and ordnance can change during the launcher firing cycle. Therefore, consideration should be made on the progression of ordnance through its launcher, and the potential for EM coupling throughout that progression.

4.4.1.2.6 Immediate Post-Launch. The ordnance configuration for the immediate post-launch phase can be very complex and difficult to replicate or simulate during HERO testing. This phase typically applies to launched ordnance and addresses the situation where the ordnance item has been physically separated from the platform or system (for example, launched, ejected) until it reaches an explosive safe separation distance (SSD) from the host with regard to the actuation of its explosives, pyrotechnics, or propellants. The immediate post-launch configuration is usually very different from the pre-launch configuration in that the item has undergone many changes in its physical state. For example, some EIDs are no longer a concern because they have been initiated, firing circuits have been altered and may be energized, and additional apertures may have been created by the jettison of components/sections or the extension of wings/fins. A detailed assessment should be performed to properly design the test configuration and setup.

4.4.1.3 Ancillary Equipment. Ancillary equipment may be necessary in a number of the S4 phases. For example, when planning for tests of the transportation/storage phase, one should consider all possible types of shipping containers, especially those that are intended to provide shielding protection. If the container is non-metal and expected to provide protection, then it may be required for testing. If the container is all-metal with proper joints, seams, EMI gaskets, and so forth, the container will not require testing. In some cases, the containerized configuration includes provisions for built-in tests or external connections to test equipment. Such configurations require testing with the test equipment connected to the ordnance and both may need to be functional during testing. In addition, container-handling equipment, cranes, or weapon lifting equipment should be considered for possible “antenna” effects. Such equipment should be included in the test procedures for the transportation/storage phase if it is determined that there is a potential impact on EID response.

Any special tools or equipment used for purposes of assembly/disassembly should be included in the test procedures because of the influence such equipment may have on the frequencies and levels of EM energy that couple into the EIDs. For the same reason, diagnostic gear (for example, continuity checkers, external test equipment needed to exercise built-in tests, special tools, and loading equipment) should be included in procedures for testing handling/loading configurations, if it is determined that there is a potential impact on EID response.

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Loading procedures often require the use of cranes and scissors lifts, as well as other equipment with a demonstrated influence on EID coupling. Platform-loaded configurations also may be subject to diagnostic evaluations requiring external connections and the use of special tools. Special test fixtures may be necessary to support the ordnance in a simulated post-launch configuration. Such fixtures should be designed to be non-perturbing and incorporate the use of electrically non-conductive materials.

4.4.2 Operating Procedures. Experience has shown that procedures used to handle, load/download, diagnose, or otherwise operate the ordnance can influence the response of the EIDs. Typically, ordnance operating procedures are specified for assembly/disassembly, handling/loading, and transportation/storage configurations. It is extremely important that these same procedures be replicated during the HERO test during item exposure to the test EME. Due to the fact that handling and loading operations can have serious safety consequences, qualified personnel should be available to perform these procedures. Ordnance operating procedures may include both standard and special, non-standard operating procedures. NOTE: Unauthorized/ Non-standard procedures performed during testing should be noted but not considered when providing recommendation for HERO classification.

4.4.2.1 Standard Operating Procedures (SOPs). Ordnance SOPs are defined in technical manuals, orders, and publications and should be used to develop HERO test procedures. However, when not available, other documentation, such as operator's guides, manufacturer's procedures, and program office guidance should be obtained and approved for use. When the SOPs become available, an assessment should be performed to determine the impact of any procedural differences on the test results.

4.4.2.2 Non-Standard (Not Specifically Authorized) Operating Procedures. Test results have demonstrated that not specifically authorized, but highly likely sequences of operation, such as touching cables or connector pins with hands or incidental contact with a grounding cable or tool, can couple sufficient energy into the device to cause a safety or reliability problem. Test preparation should include the analysis and evaluation of these possibilities and ensure that such non-standard procedures are included in the test plan to assess their impact on the susceptibility of the ordnance item's EIDs to such procedures. Methods to mitigate these susceptibilities should be included in the test report (see paragraph 8). NOTE: Unauthorized/Non-standard procedures performed during testing should be noted but not considered when providing recommendation for HERO classification.

4.4.3 Expected EME. All applicable EME levels should be defined for each S4 phase. In general, it is likely that the EME levels will not be the same for all phases. For example, ordnance items with shipboard applications are likely to encounter much lower EME levels during the handling/loading phase on the flight deck as opposed to the platform/system-loaded phase where the item will encounter much higher levels due to the exposure to radar main beam illumination. When test EME levels are being identified, it is important to have knowledge of the actual EME levels expected for each S4 phase.

4.5 Test EME Levels. MIL-STD-464 defines the HERO EME levels in terms of frequency and field strength levels and is applicable for all HERO tests. Guidance for tailoring the EME levels is included in the standard's appendix and MIL-HDBK-235. In order to receive

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a HERO classification of "HERO SAFE ORDNANCE" at the AUR or appropriate assembly level, the ordnance SUT should be evaluated against, and be in compliance with the HERO EME levels found in MIL-STD-464. When electromagnetic radiation (EMR) sources associated with the ordnance system are capable of producing fields in excess of those specified in MIL-STD-464, these higher fields are used for the ordnance evaluation.

4.5.1 EME Parameters. There are many key parameters that significantly impact ordnance response: test frequencies, field strength levels, illumination angles, polarization, and modulation. Each of these parameters is addressed in the following paragraphs.

4.5.1.1 Test Frequencies. There are two methods of establishing the EME test frequencies, the swept frequency technique and the discrete frequency technique. Swept, or continuous, frequency testing is the preferred technique from the standpoint that the test item is exposed to all frequencies within a particular frequency band; this technique improves the probability of stimulating the ordnance at resonant frequencies, where responses are greatest. If the required field strength levels are relatively low and the test facility is capable and authorized to conduct swept frequency testing, this is an excellent way to conduct HERO tests. Reverberation chambers are a highly effective means of performing swept or small-incremental frequency testing but are not sufficient for standalone HERO certification testing because of their intrinsic characteristics (see 4.3.6.6 and 4.6.3). Frequencies can be changed quickly and automatically if the RF source is computer-controlled. Even if there are significant limitations on the field strengths that can be generated, swept or continuous frequency testing may uncover resonances that would otherwise not be detected in discrete frequency testing. The value of such tests is to identify susceptible frequencies at which discrete, high-level (MIL-STD-464) testing can then be concentrated.

However, it is generally not possible to generate the required criteria of MIL-STD-464 when sweeping frequencies because of limitations in equipment or frequency authorization at OATS. But again, practical limitations in the time required to tune high-power sources generally preclude testing at finely spaced frequencies. HERO testing normally uses the discrete frequency technique in lieu of swept or fine incremental frequency testing. Here the challenge is to generate a sufficient number of discrete frequencies to characterize EID response within certain acceptable error margins.

There are practical limits on the amount of time and equipment capabilities available to capture the maximum responses. TABLE III should be used as a representative baseline for HERO testing. Test EMEs from each frequency band of TABLE III should be evaluated or otherwise should be determined to be low risk, through analysis, to satisfy the Joint HERO requirements.

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TABLE III. Representative HERO Test Frequencies in MHz

Frequency Range	Test Frequency
.01 - 2	0.15
	1
2 - 30	2.00
	2.30
	2.64
	3.03
	3.48
	4.00
	4.59
	5.28
	6.07
	6.97
	8.00
	9.19
	10.56
	12.13
	13.93
	16.00
	18.38
	21.11
	24.25
	27.86

Frequency Range	Test Frequency
30 - 150	32
	34
	38
	46
	55
	57
	63
	70
	80
	85
	95
	110
	118
	136
	140
150 - 225	150
	174
	204
	225
225 - 400	230
	264
	275
	290
	303
	350
	385

Frequency Range	Test Frequency
400 - 700	425
	500
	577
	630
700 - 790	700
	760
790 - 1000	800
	850
	910
1000-2000	1280
	1500
	1750
	1900
2000-2700	2250
	2400
2700-3600	2750
	3000
	3312
3600-4000	3700
	3900
4000-5400	4400
	4900
5400-5900	5500
	5650
5900-6000	5950

Frequency Range	Test Frequency
6000-7900	6500
	7000
	7563
7900-8000	7950
8000-8400	8200
8400-8500	8500
8500-11000	9000
	9500
	10000
11000-14000	11500
	12000
	13000
14000-18000	15000
	16000
	17000
18000-50000	18000
	20000
	24500
	26500
	28000
	34600
	39000
	45000

NOTES:
1. EME levels are defined in MIL-STD-464.
2. In some of the frequency ranges for "Restricted" testing, limiting the exposure of personnel through time averaging will be required to meet personnel safety requirements.

4.5.1.2 Field Strength Levels. The external EME levels for HERO found in MIL-STD-464, establish the HERO EME requirements. The standard describes two field strength categories, Unrestricted and Restricted. The Unrestricted levels in MIL-STD-464 apply for the S4 phases where personnel will not be present; that is, Transportation/Storage, Staged, Platform-loaded, and Immediate Post-launch.

To protect test personnel, it may be acceptable to conduct HERO testing at levels less than those indicated in MIL-STD-464. This practice is acceptable when personnel can potentially be exposed to EM environments that can exceed the maximum permissible exposure (MPE) and induced/contact current (I/CC) limits (refer to DoDI 6055.11). These limits can be exceeded through exposure of RF and/or through contact with conductive surfaces that are exposed to RF.

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When the test levels are decreased it is essential the EMEs remain within the acceptable linear extrapolation limits (see 7.7).

During the planning phase, an evaluation of the ordnance design should be conducted to determine if it could respond to peak, as well as average EME levels. This evaluation generally depends on the response time of the EID and the energized firing circuits. Initiating devices and circuits with very fast response times may respond to peak field strengths and should be tested to the peak EME levels; devices and circuits that respond only to average field strengths need only be tested to the average field strengths. Energized firing circuits and EIDs with relatively fast response times (for example, conductive composition electric primers) are sometimes referred to as "peak power sensitive;" in contrast, the relatively slow bridgewire-type EIDs cannot respond to peak environments and are referred to as "average power sensitive" devices. It is important to determine if EID firing circuits can be energized in any phase of the S4 evolution, from stockpile to the point of reaching an SSD. If testing the powered-on state is determined to be required, then the ordnance should be tested at both peak and average EME levels. If the system is deemed likely to respond to peak EME levels, it also will be necessary to determine appropriate modulation parameters (see 4.5.1.5).

TABLE IV summarizes the preceding discussion on EME levels, relating peak and average field requirements to EID/firing circuit response times for each of the six phases of the S4.

TABLE IV. HERO EME Test Levels

S4 Phases	EME Levels[1]	
	Non-energized firing circuits or slow-responding EIDs	**Energized firing circuits or fast-responding EIDs**
Transportation/Storage	Unrestricted average levels	Unrestricted peak levels[2,3]
Assembly/Disassembly	Restricted average levels	Restricted peak levels
Staged	Unrestricted average levels	Unrestricted peak levels[2,3]
Loading/Unloading	Restricted average levels	Restricted peak levels[4]
Platform-loaded	Unrestricted average levels	Unrestricted peak levels[2]
Immediate Post-launch	Unrestricted average levels	Unrestricted peak levels[2]

NOTES:
1. Applicable field intensity levels are specified in HERO Test EME in MIL-STD-464.
2. Unrestricted peak levels should be used unless tailored environments have been developed.
3. Applies to fast-responding EIDs only.
4. Some firing circuits may be energized during the loading/unloading sequence in order to accomplish pre- and post-loading diagnostic procedures.

4.5.1.3 Illumination Angles. The angle of the incident EM field has a pronounced impact on the amount of energy that couples into the EIDs. Ideally, the ordnance should be illuminated from all angles to cover all coupling paths (seams, apertures, and other discontinuities). However, limitations in equipment or test time may preclude such a complete illumination. As a consequence, the ordnance is illuminated only at discrete angles. In these instances, modeling, engineering analysis, or previous experience with similar ordnance may be useful for predicting the illumination angles expected to induce maximum coupling. Part of this prediction should consider all of the SUT seams, apertures and other discontinuities along with

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maximum coupling frequencies relative to SUT size. FIGURE 7 illustrates various illumination angles that can be used during testing.

The method of varying the illumination angle generally depends on the type of transmitting antenna being used. For fixed antennas, such as the larger antennas used at HF and VHF frequencies, the ordnance may need to be re-positioned with respect to the antenna. This can be done manually or with the use of a turntable. The use of a turntable facilitates 360-degree azimuth illumination, but requires that the declination angle of highly directive transmitting antennas be changed periodically and the azimuth rotation repeated. This method may require changing the height of the antenna. Higher-gain, mobile antennas are highly recommended for testing above the HF band, as they can be readily moved about the SUT and adjusted for different declination angles. The use of portable antennas facilitates more thorough illumination, especially when high directivities make illumination a more critical parameter. Additional guidance for selecting discrete illumination angles is provided in multiple paragraphs within 6.5.

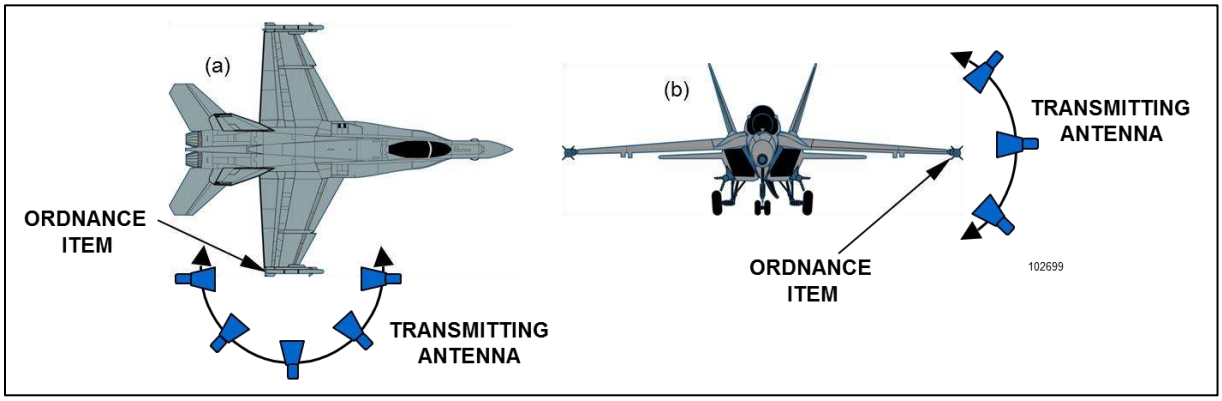


FIGURE 7. Example of Illumination Angles

No matter what methods are used to change the illumination angle, it should be remembered that these changes can have a pronounced effect on the magnitude of current induced into the EIDs. If illumination is maintained during changes in the angle of incident radiation, sufficient time should be allowed for the EID and EID instrumentation to respond fully to the changes in the induced current.

4.5.1.4 Polarization. Polarization of the test environment should be consistent with the anticipated field conditions for the ordnance. For example, a vertically polarized antenna should be utilized in the 2- to 30-megahertz (MHz) band since antennas that operate in this range are typically vertically oriented. Both horizontal and vertical polarization should be utilized above 30 MHz.

4.5.1.5 Modulation. For most HERO testing, the average EME level is the primary concern. This is because the majority of bridgewire EIDs have relatively long response times (tens of milliseconds). Most EIDs are not very responsive to the type of short duration, peak EME levels associated with common modulation formats. In such cases, the magnitude of the induced EID current is no greater for a modulated EME than for a CW EME, assuming the same average levels for both. Modulation may be used to conduct the test, but the EID response should be defined in terms of average field strength levels. However, there are some EIDs that

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exhibit extremely fast response times (fractions of a millisecond). Conductive composition primers are an example of such devices. In these instances, the EID may have a response time fast enough to react to a single pulse or, as in thermal stacking, gradually respond over time to repeated pulses, as in a pulse train. If it is determined that the EID can respond to peak EME levels for single pulses or from thermal stacking, the test EME should include the peak levels specified in MIL-STD-464. In addition, energized EID firing circuits should be assumed to have the potential to respond instantaneously, within a fraction of a microsecond or less, and also should be tested to the HERO peak EME levels specified in MIL-STD-464.

The requirement for testing to peak level EME levels implies the need to choose the modulation type; for example, pulse modulation (PM), amplitude modulation (AM), frequency modulation (FM), and associated modulation parameters (pulse width, PRF, tone frequency, frequency deviation, and so forth). In general, the selection of modulation parameters should be chosen to represent the environments expected to be encountered during the ordnance S4 and maximize the response of the EIDs or EID firing circuits (see 6.5 for additional guidance).

4.6 Selection of a Test Facility. A facility to test the system should be chosen such that the objectives of the test can be adequately met. In choosing a test facility, the following questions should be considered:

a. Does the test laboratory have RF generating equipment capable of generating the test EME needed for the planned system evaluation?

b. Is the size of the laboratory test area(s) large enough to accommodate the system, and its required support equipment, to be tested?

c. What equipment is available to support the system?

d. What is the electrical power that is available to support the system and test support equipment?

e. For systems that require fuel sources (that is, aircraft or vehicles), will the fuel sources be readily available?

f. What facility restrictions exist that may impact the operation of the system (for example, safety, hazardous material (HAZMAT), and so forth)?

g. If the system/program/test data will be classified/controlled, what is the capability of the laboratory/facility to secure these items? What is the general security capability of the test laboratory and its facilities?

h. What must be done to ship the system and required support equipment to the laboratory? Is special transport required for shipping?

i. Where is/are the laboratory(ies)/facility(ies) located?

j. What is/are the cost/fees that the test laboratory will charge to conduct testing?

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k. Is the test laboratory a Government facility with test and evaluation authority? This can be cost effective for potential future applications of the system, as this could avoid the need for a future retest.

There are generally three types of test facilities used for evaluation of systems in their external RF EMEs: OATS, anechoic chamber, and reverberation chamber. A test laboratory may have one or more of these types of test facilities available for use. Such a test laboratory does offer some flexibility in testing; however, if more than one type of facility is being considered to support testing, the logistics of moving the SUT/test setup from one facility to another facility will have to be considered.

4.6.1 OATS Facility. An OATS facility is simply an outdoor area designated for testing. The area is generally unenclosed (except, perhaps, by security fencing) and may have a conductive ground plane to provide for more efficient transmissions at lower frequency EMEs and/or to provide proper equipment grounding points. The facility may have built-in power availability or may have power supplied to the facility by mobile power generators. An example of a typical ground plane facility is shown in FIGURE 8.



FIGURE 8. Typical OATS test facility with ground plane

The advantages of using an OATS are the following:

a. Generally able to accommodate large systems for testing.

b. EME reflections are minimized.

c. EMEs are generally predictable and therefore easy to characterize.

d. Minimal restrictions on EME generation based on physical constraints (for example, boundary conditions, resonant cavity effects, very low Q).

e. Hazards such as fire, engine noise and exhaust (that is, from aircraft, vehicles, and power carts) are easier to mitigate than hazards in enclosed facilities.

f. Less physical restrictions to mitigate for testing RF based systems (for example, radar, global positioning system (GPS), transceivers).

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g. Because testing is conducted at specific antenna locations and EME polarizations, EME susceptibility POEs are easier to identify.

The disadvantages in using an OATS are the following:

a. Exposure to weather; severe weather can add hazards and delays to testing.

b. Legal restrictions on EME generation (that is, Federal Communications Commission (FCC) regulations, National Telecommunications and Information Administration (NTIA) regulations, or laboratory policies/restrictions for open-air transmission of an EME).

c. Potential in causing unintentional EMI in equipment and systems that are not part of testing and/or belong to other activities.

d. More difficult to control ambient EME than in enclosed facilities.

e. Security concerns: the test item(s) and support equipment will be out in the open and potentially within view of unauthorized personnel and/or surveillance aircraft/spacecraft.

f. If the system is large, a test may require a large number of test positions per EME in order to sufficiently characterize the system. A large number of test positions will likely increase test time, cost, and would result in the analysis of a large amount of data.

g. Increased chance of inadvertent personnel exposure to the EME.

h. Potential ground reflections off the ground plane that could affect calibration or system exposure.

4.6.2 Anechoic Chamber. The anechoic chamber is a type of shielded enclosure lined with radar absorbing material (RAM) to reduce or eliminate RF reflections. The RAM in the enclosure comes in different shapes (usually pyramidal) and sizes designed to efficiently absorb EM energy in a certain frequency range. Multiple types/sizes of absorber material are generally used in anechoic chambers to increase the effective operational frequency range of the chamber. RAM is also rated as to how much EME power it can absorb/handle, making it imperative that the test director confirms that the anechoic chamber can handle testing at the field levels required for the system's EME evaluation. FIGURE 9 shows a typical anechoic chamber.

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FIGURE 9. Typical anechoic chamber

The advantages in using an anechoic chamber are the following:

a. No exposure to weather, as testing is conducted indoors.

b. No restrictions associated with spectrum usage.

c. No unintentional EMI in equipment and systems that are not part of testing and/or belong to other activities (that is, equipment in nearby buildings).

d. Better control of ambient EME as testing is isolated from the external environment.

e. Eliminates exposure of non-test personnel to the test EME.

f. EME susceptibility POEs may be easier to identify based on specific antenna locations and EME polarizations.

g. Security controls may be minimized since access to the test location is better controlled.

h. EME reflections are minimized.

i. EMEs are generally predictable and therefore easy to characterize.

The disadvantages in using an anechoic chamber are the following:

a. Chamber dimensions may limit the size of systems that can be tested.

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b. If the system is significantly large, a test may require a large number of test positions per EME in order to sufficiently characterize the system. A large number of test positions would add to test time, cost, and would result in the analysis of a large amount of data.

c. May be restricted from testing of lower frequency EMEs due to the physical constraints of the chamber and/or absorber material.

d. Hazards such as fire, engine noise and exhaust (that is, from aircraft, vehicles, and power carts) may require more mitigation/work-around procedures than hazards in OATS facilities. Systems/equipment that contains/uses hazardous components (e.g., flammable fuels, explosives, lithium batteries) may be severely restricted for use, or prohibited.

4.6.3 Reverberation Chamber. The reverberation chamber is a highly reflective (high Q) shielded enclosure that is excited by an RF source at frequencies where the chamber dimensions are large compared to excitation wavelength. The simultaneously existing modes within the chamber are stirred by a field-perturbing device referred to as a paddle-wheel tuner. As the tuner assembly is rotated by means of a stepper motor, the complex standing-wave patterns are altered significantly and the SUT is exposed to randomized fields from all directions and polarizations. Because of these reasons, the reverberation chamber is an effective tool and can be used as a supplement to traditional open-air and anechoic chamber test techniques. FIGURE 10 shows a typical reverberation chamber.

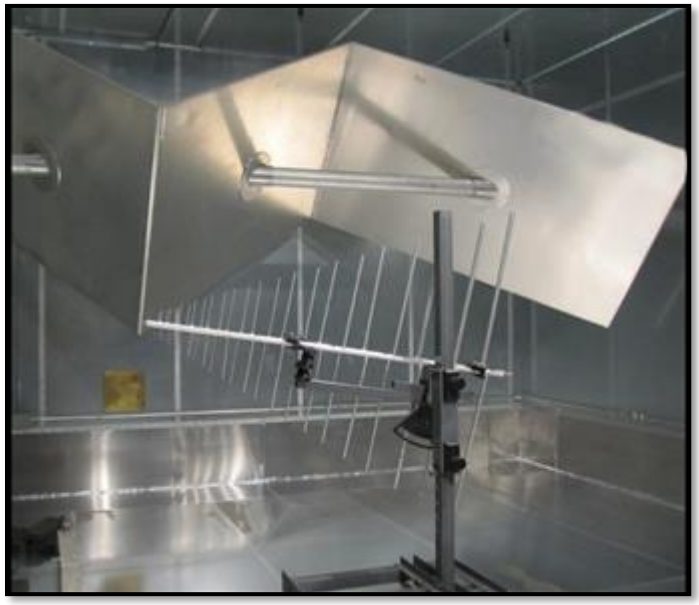


FIGURE 10. Typical reverberation chamber

The usable frequency range of a reverberation chamber, particularly the lowest usable frequency (LUF), should be understood and taken into consideration as part of the test planning process. In order to ensure that the EME inside the chamber is statistically uniform and isotropic over the course of a complete tuner rotation, the chamber must have a sufficient mode density at its LUF.

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The LUF can be determined by using equation 4-3 to solve for the frequency where the number of modes (N) is equal to 60.

$$LUF = c\left(\frac{3N}{8\pi abd}\right)^{1/3} \quad (4\text{-}3)$$

where:

a, *b*, and *d* are the chamber internal dimensions in meters
N is the number of modes in the cavity ($N \geq 60$)
LUF is the lowest usable frequency in Hz
c is the speed of light (3×10^8 m/s)

There are two modes of operation for a reverberation chamber, defined by the way the paddle-wheel tuner is rotated: mode-stirred and mode-tuned. In a mode-stirred operation, the chamber is excited at a single frequency while the paddle-wheel tuner is continuously rotated over 360 degrees. Mode-stirred operation is often not the preferred mode of operation for external RF EME tests, since the number of samples (for example, the response intervals of the SUT) may vary significantly depending on the rotation rate of the tuners and the response time of the equipment. Furthermore, at higher frequencies, where the modal density of the chamber is higher, the rate of change for the modal structure in the chamber, which directly affects the EME levels impinging on the SUT, can change very quickly for moderate tuner rotation rates.

In a mode-tuned operation, the paddle-wheel tuners are moved in statistically determined increments and held stationary while a measurement is made at each position. A statistically uniform and isotropic environment is achieved by indexing the tuners through a 360-degree rotation for a given input frequency and power level.

The advantages and disadvantages in using a reverberation chamber for testing are basically the same as the advantages and disadvantages in using anechoic chambers for testing, with the following exceptions:

The advantages in using a reverberation chamber are the following:

a. The high Q characteristic allows significant field levels to be generated with relatively small input power.

b. Because the SUT is exposed to randomized fields from all directions and polarizations by the rotation of the paddle wheel tuners, exposure of the entire SUT to the proper test EME reference field levels at any time during a test run is statistically assured and uniform.

The disadvantages in using a reverberation chamber are the following:

a. Because the SUT is exposed to randomized fields from all directions and polarizations by the rotation of the paddle wheel tuners, determining EME POEs into the SUT may be difficult and require additional investigation, as the EME polarization and direction are obscured.

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b. Chamber calibration required prior to conducting test.

c. Frequency change is limited.

d. Personnel may not be allowed to be located within the test chamber during testing.

4.7 EID Instrumentation. There are essentially two test methods used for HERO evaluations, instrumented and non-instrumented. Instrumented tests are preferred over non-instrumented (go/no-go type) tests because instrumented tests provide a detailed quantification of the ordnance item's susceptibility to RF energy. Non-instrumented tests, if conducted properly, require a significant number of test items and test time to complete the evaluation. Otherwise, a non-instrumented test that uses very few test items or poor test techniques results in insufficient statistical data needed to determine compliance.

For the purposes of this handbook, this discussion will concentrate on instrumented testing and will provide a limited discussion on the non-instrumented testing in 4.7.5. The following paragraphs provide discussions on the various types of instrumentation systems used in ordnance testing, characteristics of an instrumentation system, the selection rationale, and other types of instrumentation used for unique non-bridgewire EIDs or firing circuits.

4.7.1 Instrumentation Characteristics. EID instrumentation should be capable of detecting and monitoring RF-induced responses of EIDs contained in an ordnance system. Yet, because HERO testing is complex and dynamic in nature, the instrumentation is also complex and very challenging. In addition, there are no military standards or specifications governing the use of EIDs contained in ordnance systems. Hence, EIDs vary significantly with respect to size, sensitivity, and thermal response time. Traditional HERO instrumentation relies on detection of the change (rise) in the temperature of the initiating element, which is directly related to the induced current. In essence, the instrumentation characteristics are as follows:

a. The sensor/transducer should be capable of detecting small changes in temperature in a device that has a small mass and, therefore, very little thermal energy associated with its temperature.

b. The instrumentation system (that is, sensor/transducer, monitoring, and recording devices) should be capable of detecting responses to short-duration pulses or stimuli.

c. The sensor/transducer should not alter the firing and EM characteristics; for example, MNFS and impedance of the EID.

d. The instrumentation system should not alter the EM characteristics of the ordnance SUT.

e. The instrumentation system should not be adversely impacted or altered by the EME.

f. The instrumentation system should be capable of operating for the duration of the test.

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g. The instrumentation package should be rugged, compact, and relatively simple to operate.

h. The instrumented EID calibration should be verified both pretest and posttest.

The following paragraphs provide discussions and guidance with respect to instrumentation sensitivity, response time, and non-perturbing effects.

4.7.1.1 Sensitivity. As general guidance, the instrumentation system should be capable of detecting a temperature rise in the EID equivalent to or less than 5 percent of the EID's MNFC. Sensitivity less than 1 percent of the MNFC is desirable and achievable with most of the instrumentation systems discussed in this paragraph for instrumentation of typical HBW EIDs. However, the most important factor concerning the instrumentation sensitivity is that it is sensitive enough to establish the required margin when the system is exposed to its expected operational EME. Another important factor is that the entire instrumentation system, including the sensor, transducer, data transmission technique, receiver, monitor, and recorder that will be used during the test should be considered when determining the system sensitivity.

4.7.1.2 Instrumentation System Response Time. The instrumentation system response time is defined as the time required for the instrumentation system to increase from 10 to 90 percent of its final value in response to a step function input. The instrumentation response time should be at least as fast as the thermal time constant of the EID being monitored. However, a response equal to or less than one-tenth of the EID thermal time constant is desired. Also, it is important to include the entire instrumentation system (for example, sensor/transducer, receiver, monitor/recorder, and analog-to-digital converter) that will be used during the test when determining the overall system response time. Integration of sensors, either in direct contact or in close proximity to a bridgewire, can result in significantly longer response times than the natural (un-instrumented) response time. The consequence of extended response times as an instrumentation artifact may limit the ability of the EID to fully respond to short duration and/or transient fields. Instrumentation response time, determined as a pre-test activity, should be compared to the EID's specified response time. In cases where the instrumented EID response time is much greater than the un-instrumented EID response time, extending the dwell time may be justified to compensate for this instrumentation limitation.

4.7.1.3 Effects of Instrumentation on SUT. It is recognized that the SUT, including the ordnance item, its launcher, aircraft, and associated firing and control cables and circuitry, when placed in an EME, can act as a receiving antenna. Its efficiency as an antenna is dependent upon many variable parameters. When these parameters satisfy certain conditions, it is possible for EM energy to be coupled into the ordnance item's EIDs.

Consequently, any alterations or modifications, such as conductive leads or housing apertures to the ordnance (for example, the introduction of an instrumentation package) may impact the antenna efficiency. This can result in either an enhancement of the RF coupling into the ordnance item or, conversely, a decrease in the RF pickup in the ordnance item's EIDs. Obviously, both of these are undesirable effects. Therefore, every effort should be made to select and implement an instrumentation package that will not impact the RF characteristics of

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the ordnance. The instrumented system should exhibit the same RF characteristics as the non-instrumented test article or, more appropriately, an in-service production item.

In addition, the instrumentation system should not be adversely affected by the EME. An all-fiber-optic-based system from sensor to receiver/recorder is preferred for HERO testing. However, the use of conductive-type instrumentation systems is acceptable, but the test engineer should be aware of the potential adverse effects and employ techniques to minimize the impact on the ordnance response. Examples of such techniques include the following:

a. Using twisted pairs with double over-braid shields terminated 360 degrees at all connector and bulkhead interfaces.

b. Running the instrumentation cables internally and in close proximity to the internal skin of the ordnance system's enclosure body (airframe).

c. Keeping the instrumentation package within the ordnance enclosure; for example, within an empty motor case.

d. Using filters, filter pins, isolation transformers, ferrites or other EMI suppressors on instrumentation cables.

e. Locating data collection components within a shielded enclosure.

f. Directing the test EMEs away from the instrumentation equipment when possible.

4.7.2 Types of Instrumentation. Instrumentation systems used in HERO programs historically have been based on measuring the temperature rise in the bridgewire of an EID, from which the equivalent induced RF current may be inferred. It has become the accepted practice for HERO testing to quantify the EID response in terms of current. Moreover, this equivalent DC current is used as a convenient point of comparison to the EID's statistical firing data that usually is given in terms of current. A typical HERO instrumentation system consists of four basic units as follows:

a. Sensor to detect an RF-induced response.

b. Transmission line to carry the sensor data to a receiver or readout device.

c. Device to translate sensor data into desirable units of measure.

d. Means of recording the data into a permanent record. FIGURE 11 and FIGURE 12 provide block diagrams of the typical HERO instrumentation systems currently used in HERO test programs.

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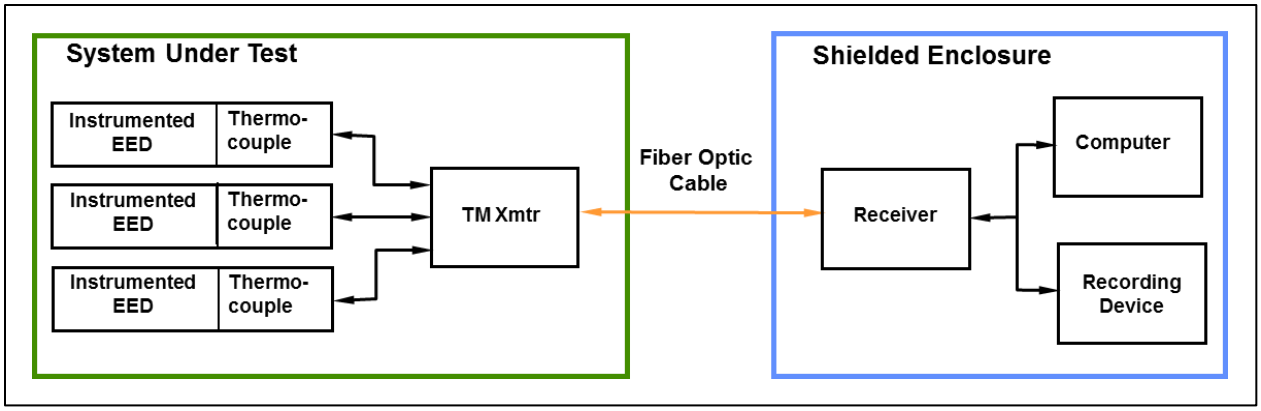


FIGURE 11. Metal-junction thermocouple instrumentation for bridgewire-type EIDs

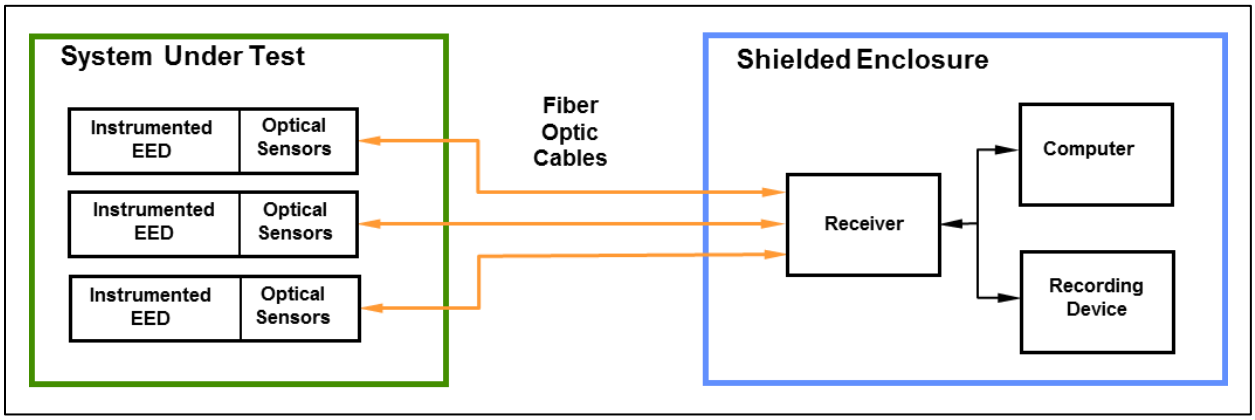


FIGURE 12. Typical optical sensor instrumentation for bridgewire-type EIDs

4.7.2.1 Sensors and Transducers. There are a variety of sensors and transducers used to detect and measure ohmic heating of bridgewire-type EIDs. These are metal-junction thermocouples, thermopiles, optical thermocouples, phosphor-based optical sensors, and vacuum thermocouple simulated EIDs. Also, in the past, some DoD activities have performed HERO tests using thermal-sensitive materials or wax-based instrumentation. However, this type of instrumentation seldom is used today.

These sensors and transducers can be divided into several categories and sub-categories to compare and contrast their inherent characteristics and advantages/disadvantages for HERO testing. FIGURE 13 provides a diagram of an optical thermocouple.

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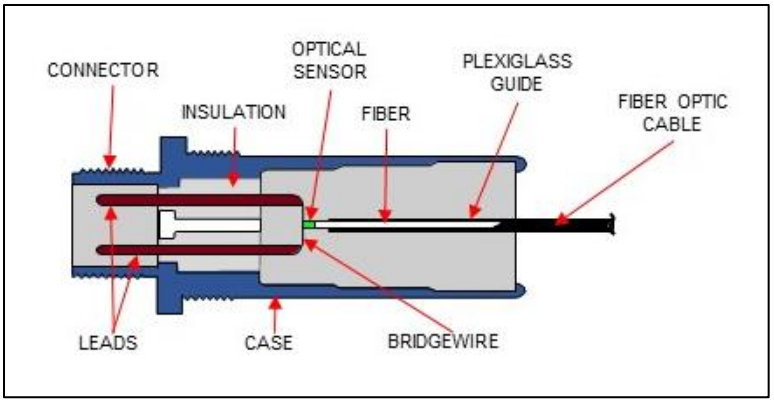


FIGURE 13. Bridgewire-type EID instrumentation

4.7.2.2 Actual EID vs. Substituted Devices. Sensors can be divided into two categories as follows:

a. Those that use the actual system EIDs (inert).

b. Those that substitute the actual EID with a simulated device.

The EM energy coupling into an EID is dependent on the RF impedance matching of the EID bridgewire/circuit and the ordnance system firing circuit. Those instrumentation systems that use the actual EID are preferred because it is not known how a simulated device compares to the actual EID's characteristic impedance, especially for frequencies above 400 MHz. If substituted devices are used in the place of actual EIDs, they must meet the same form, fit, and function as the actual EIDs and have the same characteristic impedance. Documentation showing these properties will be included in the test plans and reports.

4.7.2.3 Optical vs. Conductive-Type Sensors. Optical, sensor-based instrumentation systems are preferred over conductive-type sensor instrumentation systems because the potential impact on the ordnance system's RF characteristics is eliminated by the use of nonconductive instrumentation leads.

4.7.2.4 Sensor-to-Bridgewire Proximity. Some sensors are attached to or are in direct contact with, the EID's bridgewire. Others are positioned in proximity to the bridgewire (for example, within 0.003 inch). Non-conductive sensors (optical-based) can be placed in direct contact with or attached to the bridgewire without significantly altering its inherent electrical and EM characteristics. In addition, the position of the sensor/bridgewire is more stable and is less likely to change under environmental stress. The conductive-type sensors normally are not attached to, or in contact with, the bridgewire. Special care should be taken during the sensor installation process to ensure that the conductive sensor is positioned correctly, and that it is secured (non-conductively) in place so that its position does not change under environmental stress. However, direct contact sensors can affect the thermal inertia of a bridgewire.

4.7.2.5 Computers and Recording Devices. There are a variety of monitoring/recording devices used in HERO testing, depending on the type of sensor/transducer and data transmission device selected for the test. There are several important factors associated with the recording and

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monitoring equipment. First, it is important to ensure that the equipment is capable of displaying the results in real-time and that the results are displayed in units that easily are related to the EID criteria. In addition, it is important that these devices are included as part of the instrumentation system when determining the response time and sensitivity.

4.7.3 Bridgewire EIDs and Ordnance Instrumentation. The basic steps for ordnance instrumentation for HERO tests are to minimize both the disturbance to the EM energy created by the instrumentation package and the coupling of the RF energy to the data channels. The instrumentation package should be small and internal to the ordnance item under test. Optical telemetry techniques may be used to reduce coupling of RF energy into the signal leads. For example, no external hardwire connections and no changes to the EID circuit impedance are permitted unless the impact to the circuit can be quantified and documented. The instrumentation package itself should be mounted inside the ordnance item and, if power is required to operate the instrumentation, it will be provided through battery packs or air turbine power generators.

Normally, HERO tests are conducted using ordnance items that have been modified to remove all explosives, pyrotechnic, and combustible materials including EIDs. The explosively loaded EIDs are replaced with either instrumented versions of the inert EID or substituted devices. However, the impedance of simulated EIDs should be quantified to ensure differences can be normalized in the test data, and this analysis should be documented in the test plan and report. The instrumented EIDs are installed in the ordnance item at precisely the same location as the original EID. The instrumentation output leads are routed to the outside of the ordnance item and then routed to the instrumentation recording equipment. Implementation of HERO instrumentation (that is, disassembly/assembly of the ordnance item and installation of the replacement EIDs) may be accomplished by the ordnance item manufacturer or trained Government personnel.

The test engineer should be aware of the potential adverse effects of adding new ports for entering and exiting the external skin or internal sections of the ordnance system. Proper electromagnetic compatibility (EMC) design techniques should be used to minimize the impact on the ordnance's EM characteristics. Examples of these techniques include the following:

a. Ensure that all conductive instrumentation components remain internal to the ordnance system.

b. When entering and exiting the ordnance system, especially a section that is considered an EMI enclosure with fiber-optic cables, use existing apertures or waveguide below cut-off techniques.

c. Use shielded cables and EMI connectors when passing conductive cables through bulkheads or entering/exiting the system's outer skin.

d. Keep all conductive instrumentation cables as short as possible and ensure that these cables are located next to the internal skin of the ordnance.

e. Calibrate the instrumentation system by applying a step input of DC with a duration that is at least 10 times the thermal response time of the EID. An ammeter with at

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least one-half percent accuracy may be used to measure this pulse. The output of the sensor/transducer is measured and recorded. This output versus input is the calibration for the system. The calibration establishes the relationship of the step input into the EID bridgewire to the output parameter of the instrumentation. This instrumentation output value then can be related to the EID's MNFS established by the manufacturer or DoD agency. The calibration also will establish the instrumentation system's minimum sensitivity and dynamic range. There are three important factors to consider when calibrating the system:

1) All system components (receiver, recorder, computers, and so forth) that will be used during the test and all component settings should be calibrated.

2) The DC pulse should be inputted directly into each individual bridgewire lead or an analysis of the EID circuit should be performed to determine the amount of current applied to the particular bridgewire being monitored.

3) Calibration data should be obtained for a minimum of five points (50, 25, 10, and 5 percent of the MNFC, and just above the minimum detectable current (MDC) level). However, more calibration points will provide a better approximation of the interpolated data.

4.7.4 Instrumentation/Monitoring Firing Circuits and Electronic Devices. Instrumentation for firing circuits and electronic devices is, by its very nature, unique. Extreme care should be exercised to ensure that the instrumentation provides an accurate measurement of the voltage, current, or other response without significantly changing the test results. The primary concerns are that the shielding integrity of the circuit should not be altered by the instrumentation and that the instrumentation should not form an inadvertent antenna that may change the EM characteristics of the circuit or device under test from what would have occurred without the instrumentation installed. Instrumentation methods involving fiber-optic sensors and cables almost exclusively are used in order to achieve the goals of accurate, unperturbed measurements. (See MIL-STD-331 and U.S. Army Redstone Technical Test Center TR-RD-TE-97-01 for additional details on the instrumentation and testing of electronic circuits.)

4.7.5 Non-Instrumented Testing (Go/No-Go Type). Go/no-go type verification tests generally are not recommended because they require a significant number of test items, test time, and high-powered EME generation equipment to evaluate systems properly and should only be performed when instrumentation is not feasible. In cases where instrumentation of the device is not feasible, reasonable results can be obtained with go/no-go techniques using either a safety/reliability margin verification test or a statistically adequate sample size to ensure confidence in the results. Both methods require that the test facility have the capability to test EME levels at or above the required EME criteria levels.

4.7.5.1 Safety/Reliability Margin Verification Test. The safety/reliability margin verification method requires exposing the SUT to EME levels that are higher than the criteria EME levels by the appropriate margin defined in MIL-STD-464. The explosively loaded EIDs will have the minimum explosives needed to represent the production EID, provide visual indication of initiation, and satisfy any safety concerns. Testing at EME levels significantly

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greater than the EME criteria levels can be a problem if the test facility does not have the equipment necessary to generate the higher EME test levels. For example, if the SUT has a safety concern, then the required test EME level would have to be, per MIL-STD-464, 16.5 dB greater than the EME criteria. (For example, if the EME criteria were 1000 V/m, then the required test EME level would have to be 6,683 V/m.)

In addition, test techniques become very important, especially if the SUT contains a combination of both safety and reliability EIDs. In order to satisfy the safety margin (16.5 dB) requirements, the reliability EIDs would have to be subjected to EME test levels significantly higher than the 6 dB reliability margin specified in MIL-STD-464. A possible solution to this problem is to subject the SUT to two complete test cycles. The first test cycle would be conducted to verify the reliability EIDs and the second test cycle would be conducted to verify the safety EIDs. Prior to conducting the safety margin test, the reliability EIDs should be removed or replaced with impedance-matched dummy loads to avoid overexposure. This method would require a doubling of the time required for the verification test, a factor that should be considered when planning the testing effort.

Also, a large number of each type of EID would be required for testing. All EIDs should be replaced after each test run for a given set of conditions; that is, frequency, test EME level, operating procedures, ordnance configuration, and so forth. This replacement would eliminate the concern for altering the electrical/physical properties of the EID, leading to dudding, due to the repeated application of energy below that required for EID initiation.

4.7.5.2 Statistical Sample Size Verification Test. Basically, this type of test calls for exposure of a large number of EID samples to each set of test conditions to ensure confidence in the results. The number of fires to no-fires statistically demonstrates compliance with the required margins and confidence levels. Demonstrating compliance or meeting the requirements means that many test runs will have to be performed on a number of test samples for a given set of conditions. For example, to demonstrate compliance for the 20-mm aircraft gun system, a minimum sample size of 5,000 (primer only, otherwise inert) 20-mm rounds was required for testing. In addition, this test method requires the test facility to be capable of generating the EME criteria levels. To reduce the test time, it would be preferable if the test facility were capable of conducting tests on many samples simultaneously.

5. TEST PLAN

5.1 Introductory Information. The test plan is the foundation upon which the entire HERO test program is constructed. The hallmark of a well-written test plan is that it will allow any HERO test organization to accomplish the required evaluations with a minimum of interpretation and confusion, regardless of the organization that created it. In order to accomplish the goals of the test plan, the contents of the plan should be cogent and easily understood. In general, the test plan should contain the material covered in the following paragraphs.

5.2 System Background. The development of test criteria and procedures is dependent on having a good understanding of the system and how it is intended to be used. Therefore, the test plan should include background information for the system that identifies who

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the system is designed for, what the system is designed to do, and the platforms on which the system will be used.

5.3 Purpose. The test plan should clearly identify the reason(s) for conducting the test and the criteria to which the system is being evaluated. It should also include references to any standards, specifications, or requirements that are applicable.

5.4 Scope. The objectives of the test should be clearly identified in the test plan as should any special test considerations, limitations of the test, and/or trade-offs that have been made during test planning. It should present the overall concept of the test including the location(s), duration, schedule, number of test items, and so forth. Sufficient detail should be cited to allow the reader to clearly understand the extent of the testing.

5.5 System Test Configuration. The SUT, as well as subcomponents and any support equipment or other unique hardware planned for use during the test, should be documented in the test plan. Furthermore, details concerning the setup of the SUT at the test facility should be included as well. Particulars are described below.

5.5.1 Ordnance Item Description. The ordnance item description is derived from its specification or other source documents. It should describe the item in terms of function, technical parameters, physical characteristics, mission, and S4. If the ordnance item consists of several major components or sections, identify these and describe each. If the test item differs in major areas from previously tested items, describe the differences. If literature on the item exists, include appropriate extracts from the descriptive literature and cite the reference number or title. Ensure that the manufacturer's performance claims are not included as facts in the description. Listing physical characteristics or specification requirements rather than performance claims is advisable. The description of the ordnance item should permit a full understanding of the item's function and intended mission. Include diagrams, line drawings, or photographs. Identify all applicable launch platforms/systems and describe each of the item's S4 phase configurations.

5.5.2 Ordnance Item Mission. A description of the mission of the ordnance item is needed to understand how the item will be used throughout its service life and thus becomes the basis for determining appropriate HERO test procedures. This is the basis for selecting which S4 configurations should be tested. Further-more, thorough knowledge of the mission is helpful in selecting the test EME and determining appropriate procedures for exercising the ordnance SUT. The detailed description is required to ensure that all the appropriate hardware configurations and procedures are addressed, and also to ensure that any additional equipment required is provided for assurance of thorough HERO test data collection.

5.5.3 Description of Test Item's S4 Phases. All of the test item's S4 phases should be identified. The particular phases to be tested should be decided upon prior to writing the test plan to ensure that the entire life-cycle sequence is understood, and that the proper ordnance test configurations are described. This description should address the S4 in detail, provide the rationale for selecting the phases to be tested, and explain why other phases were not selected, how they will be assessed, and include the appropriate test procedures. The test procedures should be described in sufficient detail to allow the reader to understand exactly what will occur

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during the test. Test operations procedures, including international or MIL-STDs, should be referenced. Describe in detail how to conduct the test so that another test facility with knowledge in the testing of such materiel can follow the procedures and successfully conduct the test. Specify the data that will be required to conduct the test, the process for acquiring the data, and its justification. Describe how the test item will be operated or exercised, or the conditions to which it will be exposed in order to conduct the test.

5.5.4 Ordnance Item Test Configuration. This paragraph should describe the ordnance items procured for the test. The description should include details on the version or model to be tested; for example, production, pre-production, pilot production, tactical, exercise, and telemetry. In addition, provide a brief description on how the test item will be modified in order to accomplish the instrumentation. Describe whether the EIDs are inert or live. List all support equipment that will be required for the evaluation and give the availability, source(s), and required-by date for all equipment to be procured off-site. Also, list all hardware assets other than ground support equipment that will be requisitioned for the test. Give sources and required-by dates for all hardware assets. Hardware assets include, but are not limited to: ordnance, components, EIDs, special tools, materiel handling equipment, and special test support fixtures.

5.6 Description of the HERO Test Facility. Describe the test facility in terms of physical properties and capabilities. Identify the EM transmitters and antennas and their capability to establish the test environments in both the near and far fields. Describe the physical properties of the facility, such as open-air ground plane, anechoic chamber, screen room, and so forth. Identify any unique test equipment or fixtures that will be available for the test.

5.7 Description of the Test Environments. Briefly discuss the test EME levels. In an appendix of the test plan, include a list of test frequencies (including modulations and polarizations); the MPE levels; the applicable MIL-STD-464 HERO EME levels or the tailored field strength for each frequency; S4 phases to be tested; and the test antenna locations relative to the test item. If tailoring of the test environments has occurred, the specific rationale for the tailored levels should be detailed. This explanation will allow the evaluating activity to determine any additional tests in order to satisfy a new use or configuration of that ordnance item.

5.8 Test Instrumentation. Briefly describe the type of EID instrumentation to be utilized. Identify the types of data collection hardware and methods. The detailed description of the instrumentation system and EID data will be included.

5.9 Compliance Criteria. State the test and evaluation criteria and reference the applicable HERO EME levels in MIL-STD-464. Provide the criterion verbatim from the source, as well as the source and paragraph in parentheses following each criterion. Prior to the test, a safety engineer from the SUT's program office should determine the consequence of an inadvertent function/operation of each EID in the ordnance item. Provide the applicable firing consequence factor for each of the test item's EIDs. The HERO margins are derived from a percentage of the established EID MNFC. These margins should be included and references made to their definition.

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5.10 Test Schedule. Briefly discuss the test schedule and note that a detailed schedule is presented, which may be included as an appendix.

5.11 Safety Precautions. Reference all applicable safety precautions that are in the operational documents from which the test procedures were derived. Reference locally generated SOPs that will be in effect for the HERO test. Always include procedures to preclude personnel safety issues from EM environments during the test.

6. CONDUCTING THE TEST

6.1 Introduction. Key elements of HERO test preparation were discussed in section 4, namely, analysis of key information, determination of test requirements, and EID instrumentation. Once these steps have been addressed and a test plan has been developed (see section 5), the test activity is in a position to begin the test. This paragraph provides guidance and discusses the ideal way to conduct a HERO test. The test methodology assumes that the objective of Joint certification is to determine maximum EID responses for ordnance that is physically representative of real-world usage, and operated or exercised with defined, standardized procedures, while being subjected to the specified HERO EME levels.

Safety in HERO testing is paramount. Safety considerations begin with an awareness of the hazards. Appropriate procedures to minimize risks then should be clearly identified. The performance of certain pretest measurements, discussed in subsequent paragraphs, may be useful for determining limits on transmitter power necessary to abide by the MPE and prevent overexposure of test personnel.

Guidance concerning the generation of test EME levels presumes that the objective is to create a "worst-case" EM stimulus for the SUT. EME levels typically are defined in terms of frequency, field strength, modulation, and illumination angle, for which certain combinations will induce maximum EID responses. Each of these parameters is discussed in subsequent paragraphs.

Additional guidance also is provided to assist the test activity in determining test techniques and procedures designed to maximize EID response. Selected topics, such as non-linear effects and "over-testing," also are presented.

6.2 Safety. The generation of EM fields and testing of ordnance and ordnance systems is potentially hazardous to personnel. All personnel involved with this type of test should comply with the applicable safety standards listed below and at their test facility. Additional safety measures, such as restricting non-essential personnel from the test area, may also be observed.

6.2.1 SOP. A written SOP should be included as a part of the test plan (see section 5). The purpose of the SOP is to identify the hazards associated with HERO testing and the precautions that should be taken to minimize those hazards. In general, there are two types of hazards, those related to preparation, handling, and operation of the ordnance and its associated platform/system/ancillary equipment; and those related to the test environment.

6.2.2 Safety Briefings. A safety briefing should be conducted prior to the start of each day's test to review the SOP and familiarize all test participants with the hazards and

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precautionary measures. This briefing affords everyone an opportunity to gain a better understanding of the SOP, ask questions, and offer suggestions to improve safety. Daily refresher meetings also are recommended to reinforce the rules for safe operations or discuss safety concerns.

6.2.3 Ordnance Handling. Handling large, heavy ordnance items can be hazardous during a test event. In most cases, test laboratory personnel handling ordnance should take formal training to become certified to work on specific ordnance or operate the associated handling equipment; for example, forklifts, loading equipment, and so forth. All training/ certification requirements should be satisfied and documented. If formal certification is not required, members of the team should be instructed on proper handling/loading/operating procedures and identify any safety hazards that may exist. Ordnance handling/loading equipment should meet all individual requirements for safety certification.

6.2.4 MPE. The concern for possible overexposure exists whenever personnel are exposed to the test EME levels. Steps should be taken to ensure that personnel are not exposed to EME levels that exceed the MPEs specified in DoDI 6055.11. MPEs are specified in terms of E-field, magnetic field strengths, power densities, or I/CCs. Experience has shown that, in the HF and VHF ranges, I/CC MPEs typically impose more restrictive conditions than the radiated fields. In fact, it is not unusual to find I/CC levels approaching the I/CC MPE, even though radiated fields are well below the radiated MPE values. At radar frequencies, radiated MPEs are generally of greater concern than the I/CC.

6.2.5 Compliance with MPEs. Care should be exercised when performing the HERO testing so as not to exceed the radiated and I/CC MPEs. This can be done by limiting transmitter output power or adjusting the time during which personnel are exposed to the test EME levels. In some cases, it may be necessary to tailor the HERO requirements in MIL-STD-464. In addition to DoDI 6055.11, test personnel should also refer to IEEE C95.1, C95.1-2345, and C95.6 for more in-depth guidance on MPE limit compliance.

6.2.6 Operations Involving Host Platforms and Ancillary Equipment. A variety of hazards may exist when working with host platform/systems or ancillary equipment, such as cranes, forklifts, loading equipment, and so forth. Personnel should be made aware of these hazards and the basic rules that should be followed to avoid an accident or damage to the equipment.

6.2.7 Personal Protective Equipment (PPE). There are several types of PPE that may be required during a HERO test. Each test operation should be evaluated to determine if PPE is required to provide an extra measure of protection against specific hazards. PPEs also should be included in the SOP for the test operation. The list includes, as a minimum, hearing and eye protection, safety footwear, and protective clothing. Hearing protection is required when personnel are in the vicinity of operating aircraft engines or ground vehicles. Eye protection is required in the vicinity of rotating helicopter blades or when there is a risk of being hit by debris from live primers. Personnel working with heavy items always should wear protective footwear, such as steel-toed boots.

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6.3 HERO Test EME. The EME will be generated using communications and radar transmission equipment located on or near the test site. The typical HERO test site is an OATS, allowing for large platforms such as aircraft or ground vehicles to be included in the testing procedures. An example of an OATS is shown in FIGURE 14. FIGURE 15 provides a block diagram of the EME transmission equipment and antennas. The test EME will be measured and calibrated to the nearest conductive surface ("first skin") prior to testing at each frequency to ensure proper field strength levels. Test EME calibration should be performed using calibrated measurement equipment used within the valid calibration date. FIGURE 16 through FIGURE 18 provide the techniques used to calibrate the test EME.



FIGURE 14. OATS

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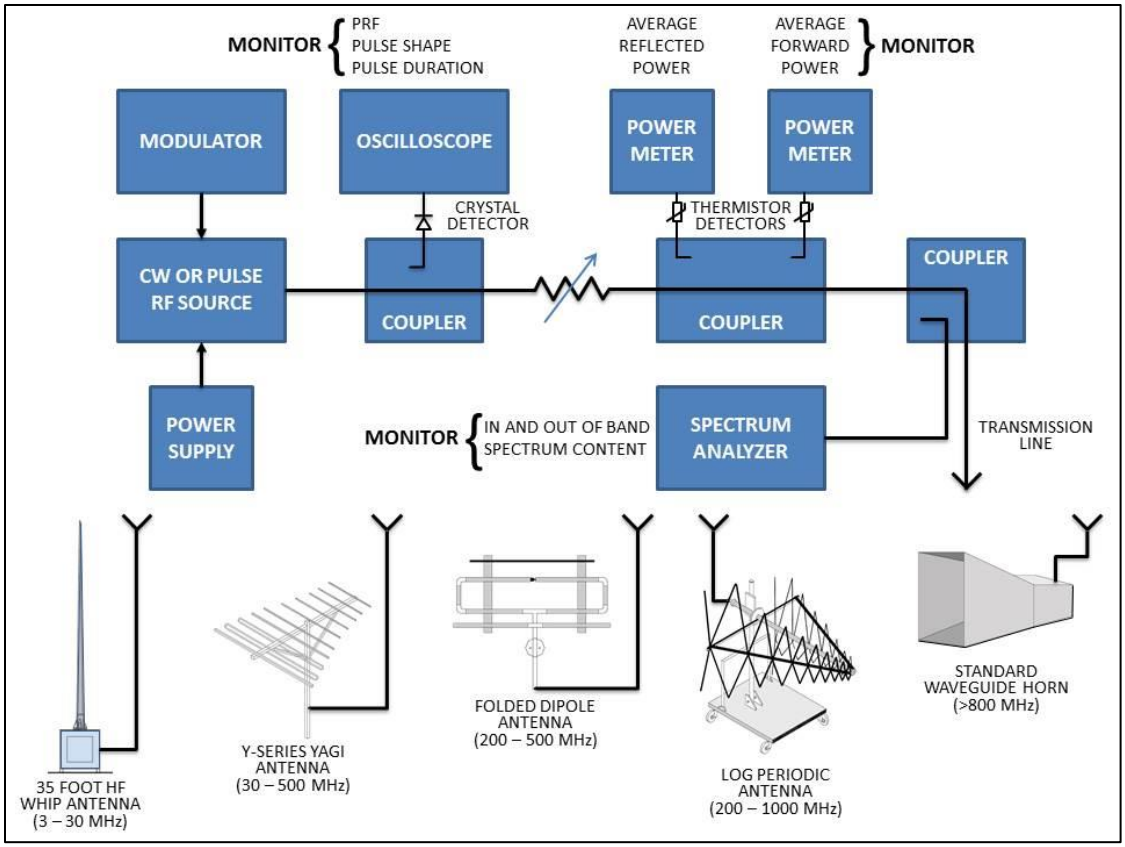


FIGURE 15. EME transmission equipment and antennas block diagram

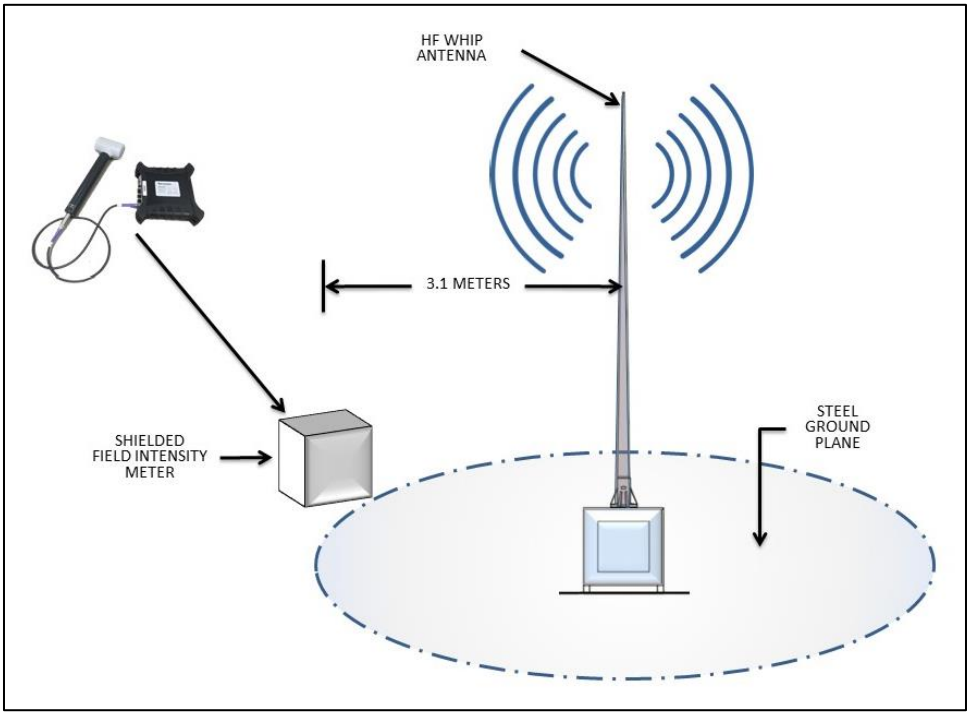


FIGURE 16. HF field calibration technique

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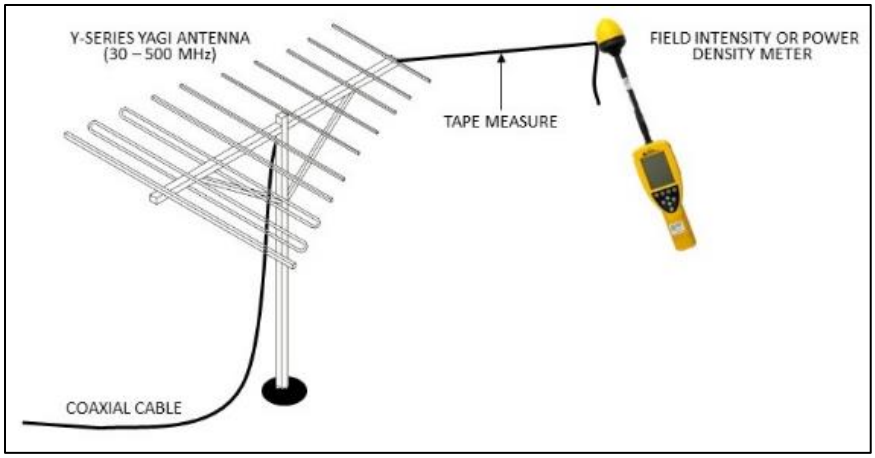


FIGURE 17. VHF/UHF range calibration technique

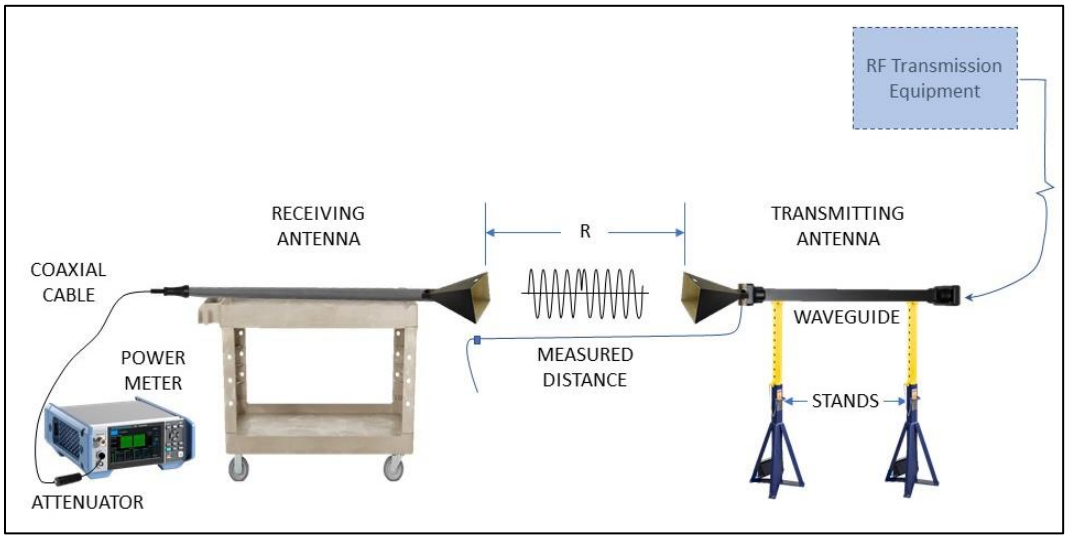


FIGURE 18. Radar range calibration technique

The EME for this test is provided in the Maximum External EME Levels for Ordnance table in MIL-STD-464. Both average and peak EME levels will be used when electronic firing circuits are powered and can initiate the EIDs, or when EIDs have relatively fast response times. The EME criteria to which ordnance systems are exposed while personnel are in contact with the ordnance are limited to the RF electromagnetic frequency (EMF) MPEs defined for controlled environments in DoDI 6055.11. In several frequency bands, the EME exceeds the continuous MPE; therefore, personnel exposure will be measured and time-averaged such that the potential power absorbed by the body does not exceed the levels defined in DoDI 6055.11. In addition, contact current and body-to-ground current measurements will be made in accordance with the IEEE C95.1-2345 and C95.3 prior to personnel making contact with the ordnance in an EME.

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6.4 General EME Calibration Procedures. This paragraph describes the EME calibration and test methodology. In general, EME testing includes Electromagnetic Radiation – Operational (EMRO), Electromagnetic Vulnerability (EMV), External Radio Frequency (RF) EME, and HERO. The scope of this paragraph excludes the following topics, which are discussed elsewhere in this document:

a. The specific characteristics of the defined EMEs that are produced in this test.

b. The identification of the equipment used to generate the EMEs.

c. The identification of the SUT, its various test configurations, and the test arrangements.

d. The placement, height, and orientation of the radiating antenna relative to the designated SUT position.

e. The procedure used to operate, monitor or inspect the SUT, or to determine the test result.

6.4.1 Free-field Calibration. In the course of testing, the SUT will be relocated to various E3 test sites that are designed to generate E-fields within a specific range of frequencies and characteristics. Free-field calibration ensures that each site will subject the SUT to the required environments by establishing the control settings that produce the required E-field level at the designated SUT ("first skin") position. The free-field calibration is conducted while the E-field is free of objects other than the E-field sensor in use. (NOTE: A free-field calibration may also be referred to as an open-field calibration.) The free-field calibration of CW, AM, and FM environments are conducted by radiating a CW signal while measuring the E-field produced at the future SUT position. (NOTE: A CW signal may also be referred to as an unmodulated signal.) The free-field calibration of a PM environment is conducted by radiating a signal with PM while measuring the E-field produced at the future SUT position. A free-field calibration is performed for each frequency of the defined EME that falls within the operational limits of the test facility and for each applicable data point (that is, for each E-field level required).

NOTE: Various transmitters and antennas are required for this test. The selection of each specific transmitter and antenna is related to the specific frequency to be radiated.

a. All test and support equipment involved in the free-field calibration will be checked for valid calibration dates. The manufacturer, models, P/Ns, serial numbers, and calibration due dates of all test equipment will be recorded along with the software versions installed.

b. The intended layout and separation distance between the radiating antenna and the SUT will be determined. The separation distance between the radiating antennas and the SUT should be sufficiently large to enclose the SUT within the radiating antenna's beamwidth 3 dB and small enough to obtain the required E-field levels at the SUT position. If it is not possible to enclose the SUT within the antenna's beamwidth of 3 dB, multiple antenna locations and/or dynamic illumination may be

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required. The arrangement of the SUT and radiating antennas will be recorded for future reference.

c. Set up the equipment that will illuminate the SUT. The radiating antenna will be positioned at one of the intended test positions.

NOTE: If an SUT is mounted to a structure or vehicle, it may be determined that the structure or vehicle intersects the beam radiated from an antenna prior to reaching the SUT. This point of intersection to the platform's nearest conductive surface is referred to as "first skin." In test arrangements with a first skin, the separation distance is measured from the antenna to the first skin rather than from the antenna to the SUT.

d. An E-field sensor will be positioned at the intended separation distance between the antenna and the SUT or first skin. The height of the sensor will be set to correspond to the vertical center of the SUT.

e. The control settings will be calibrated to produce the required E-field level at each frequency, polarization, and modulation of the defined EMEs.

1) A transmission at a required combination of frequency, polarization, and modulation will be initiated and allowed to stabilize.

2) Test personnel will adjust the transmitter controls to produce the required E-field level at the E-field sensor position.

3) Test personnel will record information to identify the selected antenna and transmitter; the orientation, height, and separation of the antenna and the E-field sensor; and the equipment settings used to produce the required E-field.

4) Terminate the RF transmission.

5) Repeat steps (1) through (4) for the next frequency, polarization, and modulation type (CW or PM) until all calibration points are completed.

f. Following the free-field calibration, the E-field sensor will be removed.

6.4.2 SUT Positioning. The SUT or SUT group will be positioned or arranged as required after the free-field calibration is complete.

a. If a number of SUTs are to be tested as a group, the calibrated separation distance will be measured from the antenna to the outside edge of the SUT group. Test personnel will ensure that all SUTs are within the 3-dB beamwidth of the radiating antenna.

b. If an SUT is mounted to a structure or vehicle, it may be determined that an antenna can be placed at the required positions and oriented in such a way that the structure or vehicle does not intersect the beam radiated from the antenna prior to reaching the

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SUT. The calibrated separation distance will be measured from the antenna to the SUT (excluding the structure or vehicle). The radiating antenna's 3-dB beamwidth, excluding structure or vehicle parts, will fully enclose the SUT. Multiple positions may be needed to cover the SUT.

c. If an SUT is mounted to a structure or vehicle, it may be determined that the structure or vehicle will effectively intersect the beam radiated from a required antenna position prior to the beam reaching the SUT. The calibrated separation distance will be measured from the antenna to the first skin of the structure or vehicle that is within the 3-dB beamwidth.

d. If a vehicle or enclosure is targeted, the calibrated separation distance will be measured from the antenna to the first skin of the structure or vehicle.

Note: The E-field levels and the induced/contact currents should be measured for safety purposes after performing Step 2 and prior to performing Step 3 of this method (see 6.4.1.e).

6.4.3 Pre-test Platform Characterization Measurements. Testing in the HF range can require a great deal of test time, especially when the ordnance item/platform/system is large and interacts strongly with the transmitting antenna. Determining the maximum EID response requires testing the item at several illumination angles at each frequency. The problem is compounded when the ordnance item can be attached at several different locations on the host platform/system. Time-consuming handling/loading procedures further exacerbate the problem. In some cases, there is simply not enough test time to complete the entire test sequence for all possible combinations of frequency, illumination angle, and ordnance attachment location. Pretest platform/system characterization has proven beneficial for identifying worst-case combinations of these parameters. If these combinations are identified prior to the test, the choice may be made to emphasize testing only at selected combinations identified as "worst-case."

6.5 Generating Test EMEs. Test activities should refer to MIL-STD-464 for applicable environment levels. In addition, for tailoring purposes, refer to MIL-HDBK-235. Any changes to EME levels require documentation in the test plan for concurrence, and in the report with respect to final assessment.

6.5.1 HF Environment (2-30 MHz). Historically, HF EMEs represent a significant HERO risk; this is because systems/EIDs are, in general, more responsive to lower frequencies, large items tend to resonate at HF, and system cables act as relatively efficient antennas at HF frequencies. Operationally, HF environments tend to be difficult to avoid because the antennas that produce them are typically omnidirectional. FIGURE 15 illustrates typical antennas, transmitters, and monitoring equipment necessary to generate test EME levels. Specific aspects of HF testing are addressed in the following paragraphs.

6.5.1.1 Field Strength Levels. The HF environment (2-30 MHz) found in the HERO table of MIL-STD-464 includes both Unrestricted and Restricted categories. The Restricted levels are applicable to ordnance used in areas (loading/unloading and assembly/disassembly)

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where personnel are exposed to the EME. It should be noted that there is a restriction on the length of time personnel can be exposed at the highest three HF test frequencies (4.5.1.2). Guidance for determining time limits at other frequencies and recommended procedures to ensure compliance with these limits can be found in 6.2.4 and 6.2.5.

6.5.1.2 Transmitting Antennas. Typically, low-gain, omnidirectional type antennas are used to generate HF test environments. In some cases, a transverse EM chamber is used; however, for Joint ordnance, a 35-foot whip is preferred. Vertical monopole or whip antennas are most common, with variations such as conical monopoles. The type of transmitting antenna used to create the HF environment can have a pronounced impact on the response of the ordnance being tested, because of strong ordnance-antenna interactions, an expected consequence of testing within the near field of the antenna. It is noted that MIL-STD-464 recommends the use of HF antennas "representative" of operational types. In near-field environments, the ordnance and antenna comprise a unique EM system, with the ordnance essentially acting as a parasitic element. Coupling efficiencies are influenced by the interactions between the ordnance (or ordnance-platform combination) and the antenna.

HF testing usually is accomplished in the near-field region. It is impractical or undesirable to conduct HF tests in the far-field for the following reasons:

a. The far-field distance is, in general, a function of wavelength and antenna dimension. There is not enough room at most facilities to accommodate the separation required for far-field illumination. In the HF range, the greatest distance is typically 1.6λ (15 meters at 32 MHz to 240 meters at 2 MHz).

b. Many facilities cannot achieve the required MIL-STD-464 EME levels at far-field distances.

c. HERO test philosophy emphasizes the importance of establishing realistic, but "worst-case" conditions (assumed, for HF, to exist in the near field).

When the EME field intensity/power density is to be measured, the measurement antenna or field probe ideally should be placed as far as possible from the source antenna as practical. While the far field is preferred it may not be practical due to amplifier and source antenna limitations.

The transfer of energy between the ordnance and the test antenna is a function of the type of antenna, the separation distance, and the orientation of the ordnance with respect to that antenna. It can, therefore, be expected that differences in types of antennas or antenna-ordnance geometry will result in differences in measured responses. MIL-STD-464 recommends test antennas be representative of the types of antennas encountered during in-service use. For example, if the item is going to be used onboard a ship, the HF test antenna should be a vertically polarized 35-foot whip antenna.

Standard facility HF antennas may not adequately represent certain types of transportable or mobile radios that will be encountered in the field. For example, man-pack HF radios produce a unique type of EME for personnel-borne ordnance carried or handled in the vicinity of these emitters. Army/Marine Corps ordnance items may be exposed to EME levels unique to the HF

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communication systems used on ground vehicle platform/systems. Evaluate these situations to determine if additional testing is warranted, that is, requiring the use of these special types of HF sources or antennas.

6.5.1.3 Field Calibration. Field calibration is the process necessary to establish the field strength at a location deemed appropriate for illuminating the ordnance or ordnance-platform/system configuration. The importance of establishing a well-defined, standardized calibration process is particularly important for near-field testing. Differences in type of antennas, ordnance location, and orientation relative to the transmitting antenna can have pronounced effects on the RF current distribution, and, hence coupling of RF energy into EIDs. See FIGURE 16 for an HF field calibration illustration.

Calibration procedures begin by tuning the transmitter to a selected frequency without the SUT in the vicinity of the transmitting antenna. An E-field measurement is made at a specified location relative to the antenna. It has been determined that the criterion for selecting this location is Service-dependent. For example, the Navy uses the "10-foot rule," based on the minimum distance allowed between the SUT, including host platform/systems, HF deck edge, and whip antennas on aircraft carriers and large-deck amphibious ships. This is considered to be the "worst-case" scenario for HF EME levels in the Navy. Thus, Navy HERO certifications are conducted by establishing a known EME level at a distance of 10 feet from the transmitting antenna. The ordnance or ordnance platform/system then is positioned with its closest point 10 feet from the transmitting antenna for all orientations of the SUT. For conical HF transmitting antennas, the separation distance is established between any element of the radiating system and the SUT. For other Services, a minimum separation distance between the radiating system and the SUT will vary depending on the type of system and how it will be fielded.

There are various types of E-field sensors used by DoD facilities to measure the HF environment. Sensors used to measure the E-field in near-field conditions should not be of a design that assumes far-field conditions.

6.5.1.4 Illumination Angles/Illumination Area. Equally spaced, symmetric illumination angles should be used for testing in the HF range. Multiple illumination angles typically are achieved by repositioning the ordnance or host platform/system with respect to the antenna. As an alternative, the SUT can be rotated on a turntable. Orientations may consist of "nose-on," or front of the platform/system, "tail-on," and each side facing the antenna. Alternatively, the test item may be illuminated from 45-degree angles with respect to the front and rear of the SUT. Experience has shown that, for HF tests involving large host platform/systems, maximum ordnance responses do not necessarily occur at the orientations where the ordnance item is closest to the antenna. FIGURE 19 depicts an SUT being tested in the nose-on orientation at a 10-foot distance from the HF transmitting antenna.

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FIGURE 19. SUT in front orientation with turret at 90° from an HF antenna

6.5.1.5 Polarization. Ideally, both vertical and horizontal polarizations should be used for HF testing; however, vertical polarization is more representative of the tactical scenario. Often changing the orientation of the SUT is a more manageable way to evaluate both polarities. When ordnance is large in more than one-dimension evaluation in two planes may be considered.

6.5.1.6 Modulations. HF modulations should include CW for testing non-energized circuits and modulated formats for testing energized firing circuits or peak power-sensitive EIDs, such as conductive composition electric primers. Modulated formats should be tested in accordance with MIL-HDBK-240-4, 6.3.5.1.

6.5.2 VHF Environment (30-225 MHz). Testing is extremely important in this range since ordnance enclosures and internal cables are of resonate lengths. Moderate-gain (6-10 dB), linear-type VHF transmitting antennas tend to illuminate all of the ordnance or ordnance-platform/system surfaces, provided the test item is located at far-field separation distances. Various aspects of VHF testing are discussed in more detail in the following paragraphs.

6.5.2.1 Field Strength Levels. Unrestricted and Restricted peak and average levels are specified in MIL-STD-464. The restricted EME levels in this frequency range reflect the maximum operational EME levels expected while personnel are executing ordnance-handling procedures (4.5.1.2).

6.5.2.2 Transmitting Antennas. Log-periodic or Yagi-type antennas commonly are used to generate VHF/UHF test environments. If testing is conducted within the near field, the same concerns apply as discussed previously for HF testing; that is, the measured responses are likely to be influenced by the type of antenna and the ordnance location/orientation. This problem

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could be eliminated if testing were accomplished at far-field separation distances. However, maintaining far-field separation distances and appropriate field strength levels may be impractical. The far-field distance is, in general, a function of wavelength and antenna dimension. In the VHF range, the greatest distance often is computed as 1.6λ. It is recommended that test activities strive to achieve the greatest separation possible, especially at the lower end of the VHF range. This amounts to a compromise between achieving desired field strength levels and maximizing the distance between the transmitting antenna and the SUT.

Standard test facility VHF antennas may not adequately represent certain types of transportable or mobile radios that will be encountered in the field. For example, man-pack HF/VHF radios produce a unique type of EME for personnel-borne ordnance carried or handled in the vicinity of these emitters. Army/Marine Corps ordnance may be exposed to EME levels unique to the VHF communication systems used on ground vehicle platform/systems. These situations should be evaluated to determine if additional testing is warranted; that is, requiring the use of these special types of VHF sources/antennas.

6.5.2.3 Field Calibration. The process for VHF field calibration is similar to that used for HF field calibration, in that it involves establishing a defined field strength at a location deemed appropriate for illuminating the ordnance or ordnance platform/system (see 6.4.1). At frequencies where far-field distances cannot be achieved, a standardized calibration process is particularly important. Differences in type of antenna, ordnance location, and orientation relative to the transmitting antenna are likely to have pronounced effects on the RF current distributions, and hence, coupling of EM energy into EIDs. FIGURE 17 illustrates the field calibration technique for VHF frequencies.

The transmitter is tuned at a selected frequency without the SUT in the vicinity of the transmitting antenna. The desired field strength should be established at a location as far away as possible from the transmitting antenna. This distance will more closely simulate most operational scenarios, minimizing near-field effects and achieving maximum illumination area. However, test EME level requirements may be difficult to achieve with available transmitter power and, thus, require relatively close distances. An E-field measurement is made at the specified location. This location is where the ordnance will be “spotted” once the test begins.

There are various types of E-field sensors used by DoD facilities to measure the VHF environment. Sensors used to measure the E-field under near-field conditions should not be of a design that assumes far-field conditions.

6.5.2.4 Illumination Angles/Illumination Area. Because the directional properties of the transmitting antennas tend to increase at higher frequencies (more narrow beamwidth), the illumination angle has a more critical effect on the ordnance response. Thus, for VHF testing, where moderate-gain antennas are used, it is highly desirable to illuminate isolated ordnance items from all angles (continuous illumination). Portable antennas and turntables can facilitate such continuous illumination. If the SUT cannot be continuously illuminated in elevation/ azimuth, the ordnance should be illuminated from at least eight angles, symmetrically selected in a 360-degree arc. If attached to a large host platform/ system, the ordnance should be illuminated at a minimum of four aspect angles in a 180-degree arc. Platform-mounted ordnance can be illuminated by re-positioning the platform/system to achieve four symmetrical

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illumination angles, with the transmitting antenna pointed directly at the ordnance item. See 4.5.1.3 for further guidance on illumination angles. FIGURE 20 depicts ordnance illumination at VHF frequencies.



FIGURE 20. Illumination of test item at VHF frequencies

6.5.2.5 Polarization. Both vertical and horizontal polarizations should be used.

6.5.2.6 Modulations. VHF modulations should include CW for testing non-energized circuits and modulated formats for testing energized firing circuits or peak power-sensitive EIDs (for example, conductive composition electric primers). AM/FM should be used to determine if modulations can interfere with firing circuit logic. Modulated formats should be tested in accordance with MIL-HDBK-240-4, 6.3.5.1.

6.5.3 Frequencies Above 225 MHz. Approximately 47 percent of the recommended HERO test frequencies are above 225 MHz (see TABLE III). For the most part, the EME levels in this range represent radar systems and are thus characterized by high peak-to-average field strength ratios. Because of the low duty cycle, it is possible that fast-responding EIDs and solid-state circuits could sustain peak power damage without any ill effects caused by the average power. In such cases, care should be taken not to expose the SUT to excessive peak levels while striving to attain specific average levels. At very low duty factors, a transmitter could produce undesirably high peak levels while attaining the desired average levels. If the SUT is considered to be peak power-sensitive, the peak field strengths may be obtained without simultaneously achieving the required average field levels. However, testing should include the required average levels.

6.5.3.1 Field Strength Levels. Unrestricted and Restricted peak and average levels are specified in MIL-STD-464. The restricted EME levels in this frequency range reflect the maximum operational EME levels expected while personnel are executing ordnance-handling procedures (4.5.1.2).

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6.5.3.2 Transmitting Antennas. For frequencies above 225 MHz, the antennas most commonly used are high-gain, aperture types, such as horns and reflectors. Far-field separation distances are achievable in most cases, thus promoting consistent results from facility to facility, despite any differences in the types of the antennas used. The far-field distance is calculated as $2D^2/\lambda$, where D is the largest dimension of the antenna. The far-field distance is, in general, a function of wavelength and antenna dimension. At frequencies above 225 MHz, the greatest distance is typically $2D^2/\lambda$, which implies separation distances less than 10 feet for the antenna types and sizes. Antenna gains of 15 dB or higher are attainable, especially at higher frequencies. Higher gains are attainable for large D/λ antennas, but without any apparent advantage in view of the corresponding increase in separation distance required to maintain far-field conditions. In fact, a notable benefit of smaller antennas is the relative ease with which they can be maneuvered about the SUT, which facilitates repositioning the antenna, in lieu of re-positioning the SUT/platform, as with the lower frequency antennas, to establish different illumination angles. It may be helpful to mount medium-to-large-size antennas on wheeled platforms/systems to further ease relocation. The use of a flexible waveguide at/near the antenna in lieu of a rigid guide also facilitates antenna repositioning and polarization changes.

6.5.3.3 Field Calibration. Field calibration at frequencies above 225 MHz is accomplished similar to frequencies in the VHF/UHF range (see 6.5.2.3). The transmitter is tuned at a selected frequency without the SUT in the field. An E-field measurement is made at a specified location relative to the antenna, where the ordnance will be placed once the test begins. FIGURE 18 illustrates an example of a field calibration technique for frequencies above 225 MHz. Field measurements can be made with field probes that read out in units of either field strength or power density. Assuming that measurements are made under far-field conditions, power density readings can be converted easily to units of field strength using the following relationship:

$$E = 61.4\sqrt{P_d} \qquad (6\text{-}1)$$

where:

E = E-field strength in V/m
P_d = power density in mW/cm^2

Measurements also can be made using standard gain receiving antennas and power meters; E is calculated as:

$$E = \frac{2.176\sqrt{\frac{P_r}{G_r}}}{\lambda} \qquad (6\text{-}2)$$

where:

P_r = received power in mW
G_r = receiving antenna numeric gain (a dimensionless ratio)
λ = wave length in meters

Any losses from cables and attenuators should be taken into account, effectively increasing the value of P_r.

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6.5.3.4 Illumination Angles/Illumination Area. At microwave frequencies, radiation patterns are highly directional, and slight changes in illumination angles can have a pronounced effect on the response of the SUT. It is, therefore, very important to illuminate the item thoroughly from all aspect angles to determine maximum EID responses. As mentioned above, antennas that are readily re-positioned about the SUT are highly recommended to achieve complete illumination. Continuous painting of the SUT with radiation is the only way to ensure that maximum responses are determined. Ideally, the item should be illuminated continuously in azimuth and elevation. Ordnance items attached to a large host platform/system should be illuminated over a 180-degree arc, directing the beam toward suspected entry points. See 6.5.1.4 for further guidance on illumination angles.

6.5.3.5 Polarization. Both vertical and horizontal polarizations should be used.

6.5.3.6 Modulations. Typically, the most severe EME levels above 225 MHz are generated by radar equipment. This is due to the extremely high peak and average power levels associated with this type of emitter. Both the choice of pulse modulation or CW formats depends on the type of radar being simulated. Surveillance and navigation radar typically are pulse-modulated, while illuminators and special-purpose radar utilize CW. Since the modulation parameters can influence the magnitude of interference to energized firing circuits or peak power-sensitive EIDs (for example, conductive composition electric primers), modulation formats should be varied to determine minimum EME thresholds. Modulated formats should be tested in accordance with MIL-HDBK-240-4, 6.3.5.1.

6.6 Maximizing Responses.

6.6.1 Illumination Angles. For highly directional EM fields, emphasis should be placed on those illumination angles expected to maximize EID response at the predicted POEs; for example, exposed wiring, enclosure discontinuities, and exposed or poorly shielded cables. In the HF range, pretest current-ground measurements and knowledge of "worst-case" HF antenna-host platform/system orientations should be taken into account. At microwave frequencies where the illumination angle is narrow, small changes can result in significantly different EID responses; thus, changes in pointing angle should be small and carefully controlled.

6.6.2 Frequency Increments. If detected responses are within 6 dB of the failure level, testing should be repeated at additional frequencies at $f_0 \pm 0.01f_0$ in an effort to find the maximum response.

6.6.3 Stimulus Dwell Time. Changes in illumination angle and polarization should not exceed the response time of the EIDs.

The optimum procedure, particularly at frequencies where the radiation pattern is highly directional, is to slowly paint the SUT, making small changes in the elevation and azimuth. This procedure should be conducted with continuous communication between the test personnel handling the transmitting antenna and those monitoring the EID responses. The illumination angle is thus controlled to emphasize those elevation/azimuth angles that maximize EID response. This procedure has two advantages: it improves the likelihood of finding the critical

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illumination angles and it ensures that the item is exposed for a sufficient time to reach maximum response.

6.6.4 Communications. It is important to maintain continuous voice communication between EID-monitoring personnel and personnel controlling the transmit antenna illumination angle. As responses are observed, monitoring personnel can advise antenna controllers on optimum angles and polarization. Additional communication is required when personnel are in the field for specified S4 phases, such as handling/loading.

6.7 Non-Linear Effects.

6.7.1 RF Arcing. RF arcs occur when two conductive objects under RF excitation are touched together or drawn apart. Therefore, arcing generally will occur in ordnance items during connection to the EID firing circuit, interruption of the firing circuit, or during any contact to the circuit by a conductive object. Under actual or test conditions, arcing will not occur at all frequencies, but will be dependent upon the particular antenna-ordnance item coupling involved. RF arcs contain energy components over a wide spectrum of frequencies. The arc output is normally an audio frequency superimposed on a dc component. The polarity of the dc signal and the relative amplitude of the audio and dc signals primarily are dependent upon the material composition of the conductive objects that form the electrodes and the RF field strength. Because of the non-linear nature of RF arcs and the fact that they generally occur in the HF and VHF frequency bands, HERO test results obtained in these bands are limited to the extrapolation levels in 7.7 or the test field strengths. It is, therefore, extremely important that all HERO tests in HF and VHF bands be conducted at the required MIL-STD-464 EME levels or at maximum available field strengths. FIGURE 21 is a photograph showing an RF arc generated during a HERO test.



FIGURE 21. RF arc generated during HERO testing

6.7.2 Energized Circuit Susceptibility. HERO assessments of EIDs in power-off, linear-firing circuits under non-arcing conditions are straightforward and subject to the test methods and instrumentation described previously in this handbook. HERO tests of non-linear, energized EID circuits, however, present unique challenges in test methods, instrumentation, interpretation of test results, and assessment of margins. There are two basic types of non-linear

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EID firing circuits, those that have power available in the circuit but should have a triggering signal applied before the EID firing circuit is activated, and those that have power and firing signal applied simultaneously. HERO tests that involve non-linear electronic EID initiating devices, therefore, need to consider the potential HERO susceptibilities in both linear and non-linear firing circuits. HERO evaluation of both the classical and EMV/External RF EME aspects of non-linear EID firing devices require an integration of HERO and EMV/External RF EME instrumentation and test techniques. The integrated method approach is necessary because neither the HERO nor EMV/External RF EME techniques alone will fully evaluate the potential susceptibilities. Margins for assessing the classical HERO response are as described previously in this handbook. Margins for active circuit upset, considered as safety-critical and mission-critical, should have at least a 6-dB margin.

As with classical HERO tests, great care should be exercised when designing tests of non-linear EID firing devices to avoid using instrumentation or techniques that will alter the response of the SUT, thereby giving erroneous test results. Due to the non-linear aspects of electronic firing circuits, it is imperative that all HERO tests involving such circuits be conducted at the required MIL-STD-464 HERO levels or at maximum available field strength levels.

6.8 Susceptibility Investigation.

6.8.1 Identifying POE. All current HERO instrumentation systems provide for real-time observation of the EID data during the test. Once the induced current in the EID of interest has been maximized by adjusting the transmit antenna aspect angles and polarization, trial and error shielding techniques in the form of aluminum foil or tape, can be applied to cables and other apertures in the ordnance in order to reduce the coupling and thereby allowing for the identification of the entry points for the EM energy.

6.8.2 HERO Fixes and Verification. Fixes for demonstrated HERO deficiencies generally fall into the categories of filtering, shielding, bonding, grounding, or a combination of all. There are many excellent design guides that discuss the subject in great detail, such as NAVSEA OD 30393 and AD 1115. It is recommended that the HERO test engineer become familiar with several EMC design guides and implement the techniques described. Such a familiarity will allow the engineer to aggressively pursue the proper fix for any and all HERO problems that may arise. Once a fix is implemented, care should be taken to retest at the previous frequencies that caused problems. Also, some fixes may simply change the resonance of the circuit, so additional frequencies above and below the failure frequencies should be tested.

7. DATA ANALYSIS

7.1 Introduction. In order to draw conclusions and make recommendations from the results of a HERO test, the test data should be presented in a format that supports the test objectives clearly and concisely. Typically, the raw test data consists of the magnitudes of induced current, voltage, or power onto the EID during the conduct of the test procedures. The raw test data in original form is insufficient for the evaluation of the test results. Other parameters, such as EME test field, EME criteria, resistance of the EID bridgewire, EID MNFS, and the firing consequence factor (FCF), should be processed with the raw test data. The results then are converted into a format that supports the conclusions and recommendations drawn from

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the test data. This paragraph details the process necessary to accomplish the data analysis. The process is basically a three-step procedure consisting of the following:

a. Converting the EID response at the EME test level into the EID response at the criteria EME level.

b. Converting the EID response at the criteria EME level into the MAE.

c. Determining a single MAE level for each frequency band of the specified criteria EME in each of the S4 phases.

7.2 Detected Responses. EID responses usually are quantified in terms of induced current (I_i). In some cases, the response may be indicated in terms of induced voltage (V_i) or absorbed power (P_i). Since the HERO margin is most often expressed with respect to the MNFC, the absorbed power measurement should be converted from power to current, which requires knowledge of the resistance of the bridgewire. Thus, the response of the EID in absorbed power can be converted into induced current by taking the square root of the ratio of the absorbed power to the resistance of the bridgewire of the EID (R_{EID}). In the case of induced voltage, the HERO margin is expressed with respect to the MNF voltage and can be determined in the same manner as the induced current. Therefore, for the purposes of this handbook, only the calculations for determinations of the induced current will be presented since the calculations for the induced voltage or absorbed power can be determined easily.

7.3 Undetected Responses. In many instances, there are no detected responses due to limitations on the test EME levels or instrumentation sensitivity. In these cases, subsequent data analysis assumes a response at the MDC level of the instrumentation system.

7.4 Compliance Criteria (Required Margins). In order to be compliant with the HERO requirements of MIL-STD-464, EIDs of an ordnance item should maintain safe and reliable margins of protection against EM-induced currents when exposed to the radiated EME levels that are expected to be encountered during its S4. In order to receive a HERO classification of "HERO SAFE ORDNANCE" at the AUR or appropriate assembly level, the ordnance or SUT should be evaluated against, and be in compliance with, HERO EME levels of MIL-STD-464. Any failures during testing will result in a classification of "HERO SUSCEPTIBLE ORDNANCE." An analysis of the HERO test data will, therefore, form the basis for determining if the ordnance item has met these margin requirements. Consequently, the test data should be analyzed for each test procedure conducted to determine if the requirements are met. These requirements are based on two possible consequences of the unintentional initiation of any EID in the ordnance.

7.4.1 Safety Firing Consequence Factors (FCFs). The 16.5- and 6-dB HERO margin values defined in MIL-STD-464 are referred to as FCFs. To avoid safety consequences, no more than 15 percent (16.5-dB margin) of the MNFC of EIDs within ordnance should be measured during any authorized test procedure.

7.4.2 Reliability FCFs. To avoid performance degradation, no more than 50 percent (6-dB margin) of the MNFC of EIDs (or 50 percent of the MNDC for EIDs) within ordnance should be measured during any authorized test procedure.

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7.5 Conversion to EID Response at Criteria EME Level. The EID response of the tested environment should be converted into the EID response at the criteria EME level. This is accomplished by calculating the EID response at the criteria induced current, which is the induced current multiplied by the ratio of the criteria EME level to the test EME level. Equation 7-1 presents this calculation. Extrapolation of the EID-induced current to the EID response at the criteria EME level is limited to the extrapolation limits detailed in 7.7.

$$I_c = I_i(E_c/E_t) \quad (7\text{-}1)$$

where:

I_c is the criteria induced current in mA
I_i is the induced current in mA
E_c is the criteria EME level in V/m
E_t is the test EME level in V/m

7.5.1 Determination of Compliance. The calculation of the EID response at the criteria level allows for determination of the compliance margin of the EID at all test frequencies. The EID response at the criteria level is compared to the EID margin level (I_{CM}) to determine criteria compliance of the EID at a given frequency. The EID compliance margin is determined as shown in equation 7-2. If the EID response at the criteria level exceeds the EID margin level, then the EID is considered to be "non-compliant" at that specific frequency. If the EID response at the criteria level is less than or equal to the EID margin level, then the EID is considered "compliant" at that specific frequency. Each frequency band then can be evaluated to determine the compliance status by evaluation of the margin status of the EID at each frequency within the individual frequency band. For ordnance that contains multiple EIDs, the compliance status of each frequency band then should consider the cumulative compliance status of all the EIDs at each frequency band.

$$I_{CM} = (FCF_d)(MNFC) \quad (7\text{-}2)$$

where:

I_{CM} is the EID compliance margin level in mA
FCF_d is the FCF as a decimal ratio (0.15 for a safety item or 0.50 for a reliability item)
$MNFC$ is the Maximum no-fire current of the EID in mA

7.5.2 Determination of Compliance Margin. The compliance margin of the EID is a dB comparison of the maximum levels of the induced currents to the allowable levels of induced currents in the EID at each test frequency. The compliance margin is determined by the following equation:

$$CM = 20\log(MNFC/I_c) \quad (7\text{-}3)$$

where:

CM is the EID compliance margin level in mA
$MNFC$ is the Maximum no-fire current of the EID
I_c is the criteria induced current in mA

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If the compliance margin of the EID is less than the FCF, then the EID is considered "non-compliant" at that specific frequency. If the compliance margin of the EID is greater than or equal to the FCF, then the EID is considered "compliant" at that specific frequency. Each frequency band then can be evaluated to determine the compliance status by evaluation of the CM of the EID at each frequency within the individual frequency band. For ordnance that contains multiple EIDs, the CM of each frequency band then should consider the cumulative CM of all the EIDs at each frequency band.

7.6 Conversion to MAE. The MAE is limited by the guidance detailed in 7.7 and determined by tests at all frequencies of interest to which the ordnance can be exposed without degrading its safety or reliability. The MAE at a given frequency is a calculated field strength at which the current induced in the EID is equal to the MNFC of that EID, reduced by the appropriate FCF. The MAE is calculated using the following equation:

$$MAE_{calculated} = E_t(I_{CM}/I_i) \text{ or } 5E_t \text{ , whichever is less} \quad (7\text{-}4)$$

where:

$MAE_{calculated}$ is the calculated maximum allowable environment in V/m
E_t is the test EME level in V/m
I_{CM} is the EID compliance margin level in mA
I_i is the EID-induced current measurement in mA

If no response is measured during testing, use the MDC as I_i.

7.7 Extrapolation Limits. At many EM test facilities, criteria EME test levels cannot be generated over all required frequency ranges. Additionally, it is desirable to determine the maximum level to which an ordnance item can be safety-certified based upon the test data in order to facilitate the safe handling of the item in higher environments, should the need arise in the future. A commonly accepted practice is to measure the response of the EIDs at the maximum field strength capability of the test facility and extrapolate those measurements to determine the MAE. In order for this extrapolation to be valid, the following applies:

a. The instrumentation should be sensitive enough so that its MDC, when increased by the ratio of the required field to the test field, is still less than the no-fire characteristics of the EID reduced by the appropriate FCF from 7.4.

b. There should be some reasonable evidence that the EM responses of the ordnance system EID circuits are linear between the measurement field and the field to which the measurements are to be extrapolated.

If both of these conditions are met, HERO test data can be gathered at levels readily available at most test facilities and extrapolated to higher levels. Extrapolation of this value should not exceed a factor of five.

7.8 Determination of MAE per Frequency Band. Upon determination of the individual MAE for each EID at each test frequency, the MAE for the individual frequency bands can be established. The MAE for each frequency band is established by reviewing the MAE of each EID at each individual frequency within the bounds of the frequency band. The

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MAE for that particular frequency band then is set at the lowest MAE of these individual MAEs. For example, assume the ordnance item tested contains two EIDs. At 18 MHz, the MAE for EID 1 is 100 V/m and the MAE for EID 2 at 4 MHz is 50 V/m. The MAE for this ordnance item in the 2- to 30-MHz frequency band should be reported as 50 V/m.

8. TEST REPORT

8.1 Introductory Information. The main purpose of the test report is to document the results of the test, present the conclusions, and convey the engineer's recommendations in a clear and concise manner.

The contents of the test report will include an updated test plan or annotation on changes made to the test plan during the test, as well as test results, a discussion of the test results, conclusions and recommendations, test data, and photographic documentation of the test item and test setup. In general, the test report should contain the material contained in the following paragraphs.

8.2 Main Body Sections. The main body of the test report will discuss the details of the test conducted. The following paragraphs describe the information that should be included and a brief description of the contents. Those paragraphs not relevant to the test may be omitted.

8.2.1 Introduction. The Introduction provides a brief synopsis of the HERO concerns within the DoD and the need for a program to address the issues. It also informs the reader as to why HERO is an important safety/reliability issue.

8.2.2 Background. The Background paragraph provides general information about the ordnance item under test and includes a brief description of the ordnance item, its mission, configuration, S4, and program sponsor.

8.2.3 Purpose. The Purpose provides the overall test objectives. This paragraph also should reference any standards, specifications, or requirements that are applicable.

8.2.4 Scope. The Scope should present the overall concept of the test including the location(s), duration, schedule, number of test items, and so forth. Sufficient detail should be cited to allow the reader to clearly understand the extent of the testing. Special test considerations, such as known limitations, should be addressed.

8.2.5 Test Results Discussion. Present, in abbreviated form, the significant findings that will give the reader an understanding of the outcome of testing without reading the entire report. Include any technical judgments that should be highlighted to the executive reader and reference test specifics in the appendix on test results. If any of the S4 phases were not tested (for example, some test configurations may be agreed upon as being worst-case over others), justification for not testing those phases must be presented. The same holds true for any deviation in scope from the approved test plan.

8.2.6 Conclusions. State the conclusion that the test item either meets or fails to meet the applicable margins of MIL-STD-464 for reliability/safety during exposure to the criteria environments in all S4 phases.

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8.2.7 Recommendations. Based on the test results, provide recommendations to the test sponsor or program office whether the test item should be considered in compliance with the requirements of MIL-STD-464 and a HERO classification. If the test item did not meet the HERO requirements of MIL-STD-464, recommend appropriate restrictions on the use of the item. In addition, list all known current versions of the ordnance item, along with all applicable platform/systems and launchers by NSN, DoDIC, NALC, or other identifiers and make recommendations as to the suitability of each to meet the requirements of MIL-STD-464. This section should also contain a HERO data sheet for each version of the ordnance item where the results of the test would be applicable.

If fixes were applied to the SUT in order to allow the test item to pass the test, these should be documented, and recommended for application into the SUT. A retest may be required with the fix applied to verify that the fix does indeed allow the SUT to be certified as HERO-compliant.

8.2.8 Appendices. The appendices should include the EID summary table, the test results table, and the HERO Data Sheet. The following tables (TABLE V through TABLE VII) are examples of what should be included.

TABLE V. EID Identification/Characteristics

EID Reference Number	Function	DoDIC	NSN	Part Number	MNFC (amps)	Resistance (ohms)	Firing Effects
A-1	Parachute Deploy Rocket	MT29	1377-01-246-5279	MK 122 Mod 0	1.0	0.9-1.3	Safety

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TABLE VI. Sample Test Results Sheet Format

SUT:						
Platform:						
Launcher:						
EID:						
Test Number:						
S4 Phase:						
MDC: (%)	**FCF$_d$:**		**MNFC: (mA)**			
EME			**RESULTS**			
464 Frequency Range (MHz)	**Test EME (V/m-rms)**	**464 Criteria EME (V/m-rms)**	**Measured IC (mA)†**	**Compliance Margin (%)†**	**Compliance Margin (dB)†**	**Calculated MAE (V/m)**
	E_t	E_c	I_i	$\left(\frac{I_i * \left(E_c/E_t\right)}{MNFC}\right) * 100$	$20\log\left(\frac{MNFC}{I_i * \left(E_c/E_t\right)}\right)$	$E_t * \left(\frac{MNFC * FCF_d}{I_i}\right)$ OR $5E_t$ whichever is less.
2-30						
30-150						

Insert all frequency ranges necessary.

18000-50000						

† = MDC used for calculations

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TABLE VII. Sample HERO Summary Datasheet

HERO Summary Datasheet			
NALC/DODIC/CRC:	**NSN:**	**MILITARY BRANCH:** Choose One	**P/N:**
ITEM NAME:			**DRAWING NUMBER:**
			TEST ACTIVITY REFERENCE:
NOMENCLATURE:			**TEST COMMAND:**
			TEST REFERENCE DATE:
			SECURITY CLASSIFICATION: (Please note that this also needs to be changed at top and bottom of the page if this datasheet is classified.) Choose One
PLATFORM:		**LAUNCHER:**	**HERO CLASSIFICATION:** **Choose One**
CONTAINER TYPE:		**CONTAINER MATERIAL:**	**EVALUATION STANDARD:** Choose One
COMMENTS:			

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Frequency (MHz)		Stockpile-to-Safe-Separation Sequence (S4) Phases					
>=	<	Transportation/ Storage	Assembly/ Disassembly	Handling/ Loading	Staged	Platform-Loaded	Immediate Post-Launch
		(V/m-rms)	(V/m-rms)	(V/m-rms)	(V/m-rms)	(V/m-rms)	(V/m-rms)
0	2						
2	30						
30	150						
150	225						
225	400						
400	700						
700	790						
790	1000						
1000	2000						
2000	2700						
2700	3600						
3600	4000						
4000	5400						
5400	5900						
5900	6000						
6000	7900						
7900	8000						
8000	8400						
8400	8500						
8500	11000						
11000	14000						
14000	18000						
18000	50000						
Consequence		Choose One	Choose One	Choose One	Choose One	Choose One	Choose One
SRAD*							

* Susceptibility Radiation Hazard (SRAD) Codes are defined in Part 7, Section 8 of this document.

9. ORDNANCE CHANGES THAT WOULD SIGNAL A RETEST

9.1 EID Changes That Require a HERO Test.

a. Lowering fire stimulus response.

b. Changing the type of transducer that converts electrical input to energetic output.

c. Changing the fit or form of the EID.

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d. Changes to EMR suppression components.

e. Relocating EID to within approximately 0.667 inches of wiring that enters or exits ordnance envelope.

9.2 Firing Circuit Changes That Require a HERO Test.

a. All non-DC firing circuits.

b. Relocating firing circuits to within approximately 0.667 inches of wiring that carries signal or enters or exits ordnance envelope.

c. Changes to EMR suppression components.

9.3 Ordnance Envelope Changes That Require a HERO Test.

a. Any added aperture or antenna.

b. Changes to corrosion protection at mating seams.

c. Changes to wiring running outside the ordnance envelope.

d. Changes to EMR absorber coatings or gaskets.

e. Relocating connectors involving ordnance subassemblies that contain an EID.

9.4 S4 Changes That Require a HERO Test.

a. Changes in authorized assembly/disassembly spaces for operations that expose EIDs or internal wiring of ordnance subassemblies that contain an EID.

b. Changes in shipping/storage container where container is intended to provide protection from EMR.

c. Change in carriage/launching platform with increased EMR induced voltages on platform.

d. A different carriage/launching platform that had not previously been evaluated.

10. DOCUMENTATION

10.1 Introduction. The documentation may be prepared in accordance with sections 5 and 8.

10.2 Purpose of Test Documentation. Test documentation verifies and validates the total efforts of a test program. Without proper documentation, a full test program could be forfeited. The test plan lays the groundwork for accomplishing the mission of HERO certification/compliance. All the minute details for conducting the test should be captured here. The substance of the HERO test program resides in adherence to the test procedures during the

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test and in the data collection process. Faithful execution of the test procedures and explicit data collection will bring the test goals into focus and lead to a successful completion of the test mission. The test report unifies the goals and objectives by summing up the test data into viable conclusions and recommendations, thus finalizing the process.

11. NOTES

11.1 Intended Use. This handbook can be used by any facility engaged in HERO testing for any of the military departments.

11.2 Supersession Data. Not applicable.

11.3 Cross-Reference of Classification and Substitutability Data. Not applicable.

11.4 Subject Term (Keyword) Listing.

E3
EID
Electromagnetic
Electromagnetic Environmental Effects
Electromagnetic Radiation Hazards
EMRH
Hazards of Electromagnetic Radiation to Ordnance
HERO
Maximum Allowable Environment
MAE
Ordnance

11.5 Identification of Changes. Not applicable.

11.6 Technical Points of Contact. Requests for additional information or assistance on this standard can be obtained from the following:

Air Force

Air Force Safety Center
HQ AFSEC/SEW
9700 G Avenue, SE
Kirtland AFB, NM 87116
(505) 846-0450
HQAFSCSEW@us.af.mil

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Army

US Army Test and Evaluation Command (HQ, ATEC)
6617 Aberdeen Blvd., Building 2202
ATTN: CSTE-OPS
Aberdeen Proving Ground, MD 21005-5001
(443) 861-9664
usarmy.apg.atec.mbx.atec-hq-g3-5@mail.mil

Army Test and Evaluation Command (ATEC)
E3 Test Division (E3 for Missile System Functions)
ATTN: TERT-ECE
Building 8978
Redstone Arsenal, AL 35898-8052
(256) 313-6970
usarmy.redstone.atec.list.rtc-jocg@mail.mil

Army Test and Evaluation Command (ATEC)
White Sands Test Center
Survivability, Vulnerability and Assessment Directorate
ATTN: TEWS-SVE (HERO)
21225 Headquarters Avenue
White Sands Missile Range, NM 88002
(575) 678-6307
usarmy.wsmr.atec.list.datts-sv-e@mail.mil

Marine Corps

Command Safety (00T)
Marine Corps Systems Command (MARCORSYSCOM)
ATTN: 00T
2200 Lester Street
Quantico, VA 22134-6050
(703) 432-4930
mcsc_safety@usmc.mil

Navy

Commanding Officer
Naval Ordnance Safety and Security Activity
ATTN: N8
3817 Strauss Ave, Suite 108
Indian Head, MD 20640-5151
(301) 744-6003
nossa_hero@us.navy.mil

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Commander, Dahlgren Division
Naval Surface Warfare Center
ATTN: B52
5389 Bronson Road, Suite 165
Dahlgren, VA 22448-5153
(540) 653-9721
DLGR_NSWC_RADHAZ.fct@navy.mil

Joint Spectrum Center

DISA/JSC
ATTN: J5
2004 Turbot Landing
Annapolis, MD 21402-5064
(410) 919-2744
disa.usna.os.list.jsc-j5-e3-engineering@mail.mil

Any information relating to Government contracts should be obtained through contracting officers.

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CONCLUDING MATERIAL

Custodians:
Army – SY
Navy – SH
Air Force – 11
DISA – DC5

Preparing Activity:
DISA (DC5)
(Project EMCS-2022-020)

Review Activities:
Army – AC, AM, AR, AT, AV, CE, CR, GL, MD, MI, PT, TE
Navy – AS, CG, EC, MC, OS, YD
Air Force – 13, 19, 22
DTRA – DS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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